

Lab Project N.º 2 - BGP ROUTING CONFIGURATIONS

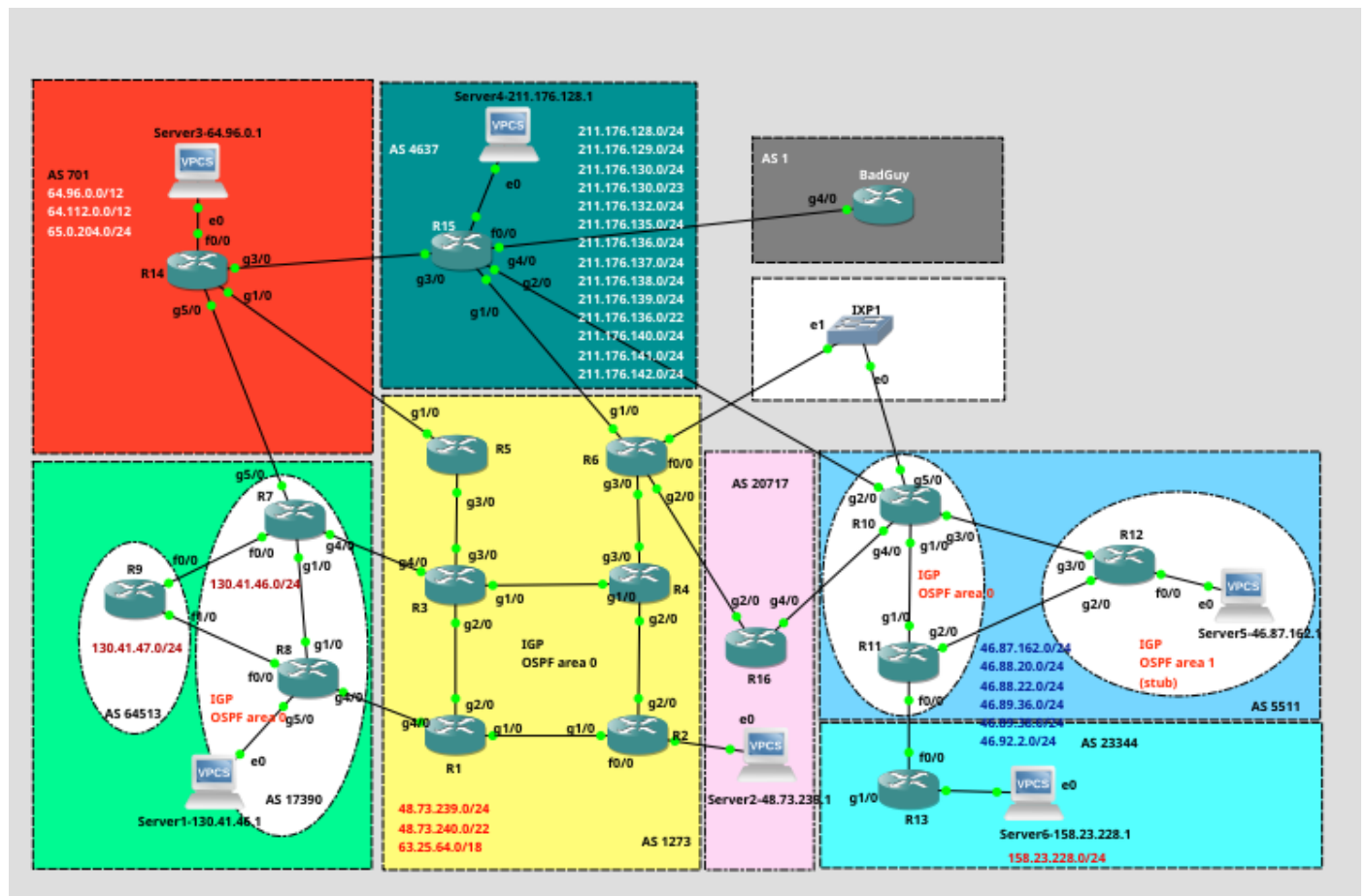
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Group 1 - Class 51D

Professor Luís Mata

luis.mata@isel.pt

- 45824 Nuno Venâncio
- 49420 André Carrilho
- 51454 Hugo Leal



Index

- Introduction
- 1 - Phase 1 - IGP Configuration
 - 1.1 - Objectives
 - 1.2 - Implementation
 - 1.3 - Test and Validation
 - 1.4 - Practical Questions
- 2 - Phase 2 - iBGP and eBGP Without Routing Policies
 - 2.1 - Objectives
 - 2.2 - Implementation
 - 2.3 - Test and Validation
 - 2.4 - Practical Questions
- 3 - Phase 3 - Routing Policies Implementation
 - 3.1 - Objectives
 - 3.2 - Implementation
 - 3.3 - Test and Validation
 - 3.4 - Practical Questions
- 4 - Phase 4 - Influence the Internet Routing
 - 4.1 - Objectives
 - 4.2 - Implementation
 - 4.3 - Test and Validation
 - 4.4 - Practical Questions
- 5 - Phase 5 - Security Practices
 - 5.1 - Objectives
 - 5.2 - Implementation
 - 5.3 - Test and Validation
 - 5.4 - Practical Questions
- Conclusion

Introduction

This project provides a comprehensive, hands-on exploration of **Border Gateway Protocol (BGP)** – the foundational protocol for routing between Autonomous Systems (ASes) on the global Internet.

In this lab, we built a realistic, multi-AS topology that simulates an Internet ecosystem, including Tier-1 and Tier-2 Internet Service Providers (ISPs), content providers, enterprise networks, and an Internet Exchange Point (IXP). The project was structured into five progressive phases, each building upon the previous to develop a complete, policy-driven, and secure inter-domain routing environment.

The core objectives of this lab were to:

1. **Establish a Stable Interior Foundation:** Configure and validate **OSPF** as the Interior Gateway Protocol (IGP) across all ASes. This included implementing a hierarchical multi-area OSPF design within AS 5511, featuring a backbone area (Area 0) and a stub area (Area 1) to optimize internal routing efficiency and scalability.
2. **Master BGP Fundamentals:** Establish internal (iBGP) and external (eBGP) BGP sessions, understand next-hop resolution, and implement scalable designs using route reflectors to avoid full-mesh iBGP requirements.
3. **Implement Routing Policies:** Use BGP attributes and filtering mechanisms to control route advertisement, influence path selection, aggregate prefixes, and enforce operational and security best practices.
4. **Enhance Network Security:** Deploy critical security measures including BGP session authentication (MD5), bogon prefix filtering, and advanced DDoS mitigation techniques like Remote Triggered Black Hole (RTBH) filtering.

This report documents our group's systematic approach to each phase. We present detailed configurations, validation tests, and thorough analyses of the practical questions. Through this project, we have gained invaluable, practical experience—from designing a robust OSPF backbone to implementing complex BGP policies—equipping us with the skills necessary to design, operate, and secure the routing infrastructure of the modern Internet.

1 - Phase 1 - IGP Configuration

1.1 Objectives

- Configure the IP addresses for all the interfaces
- Configure the OSPF as per the design rules on the ASes
- Implement the OSPF multi-area topology in AS 5511: area 0 and area 1 (stub)
- Test the private infrastructure connectivity's inside the ASes
- Test the interfaces connectivity between the ASes

1.2 Implementation

We assign IP addresses to all interfaces according to table 1 in Appendice *IP Addressing for Lab Topology*

Router example:

```
interface Loopback0
 ip address 10.1.1.1 255.255.255.255
!
interface Loopback1
 ip address 48.73.239.11 255.255.255.255
!
interface GigabitEthernet1/0
 ip address 10.1.2.1 255.255.255.252
 no shutdown
!
interface GigabitEthernet2/0
 ip address 10.1.3.1 255.255.255.252
 no shutdown
!
interface GigabitEthernet3/0
 ip address 48.73.240.1 255.255.255.252
 no shutdown
```

Server example:

```
ip 130.41.46.1 130.41.46.2 30
set pcname Server1
```

OSPF Configuration Example

AS 1273 - Vodafone and **AS 17390** - IBM

```
! All Routers -----
router ospf 1
 passive-interface Loopback0
 network 10.0.0.0 0.255.255.255 area 0
 log-adjacency-changes
```

AS 5511 - FTRSI (Orange - Worldwide IP Backbone)

```
! ROUTER 10 -----
router ospf 1
 router-id 10.10.10.10
 network 10.10.11.0 0.0.0.3 area 0    ! R11
 network 10.10.12.0 0.0.0.3 area 1    ! R12
 network 10.10.10.10 0.0.0.0 area 0    ! Lo0
 area 1 stub
 log-adjacency-changes

! ROUTER 11 -----
router ospf 1
 router-id 10.11.11.11
 network 10.10.11.0 0.0.0.3 area 0    ! R10
```

```

network 10.11.12.0 0.0.0.3 area 1      ! R12
network 10.11.11.11 0.0.0.0 area 0    ! Lo0
area 1 stub
log-adjacency-changes

! ROUTER 12 -----
router ospf 1
router-id 10.12.12.12
network 10.11.12.0 0.0.0.3 area 1      ! R11
network 10.10.12.0 0.0.0.3 area 1      ! R10
network 10.12.12.12 0.0.0.0 area 1     ! Lo0
network 46.87.162.0 0.0.0.255 area 1   ! Server5
area 1 stub
log-adjacency-changes

```

1.3 Test and Validation

Test the private infrastructure connectivity's inside the ASes

AS 1273 - Vodafone

```

! From Router 1 - Pings to all other AS routers Loopback 0 -----

R1#ping 10.2.2.2 ! Router 2 -----

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.2.2.2, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/24/72 ms

R1#ping 10.3.3.3 ! Router 3 -----

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.3.3.3, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/16/36 ms

R1#ping 10.4.4.4 ! Router 4 -----

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.4.4.4, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 20/26/40 ms

R1#ping 10.5.5.5 ! Router 5 -----

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.5.5.5, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/25/32 ms

R1#ping 10.6.6.6 ! Router 6 -----

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.6.6.6, timeout is 2 seconds:
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 24/36/52 ms

```

OSPF configuration analysis example Router 1

```

! ROUTER 1 -----
R1#sh ip ospf database

        OSPF Router with ID (48.73.239.11) (Process ID 1)

        Router Link States (Area 0)

Link ID        ADV Router    Age          Seq#          Checksum Link count
48.73.239.11   48.73.239.11   1998        0x8000000A   0x004493   3
48.73.239.22   48.73.239.22   459         0x80000008   0x004971   3
48.73.239.33   48.73.239.33   1969        0x80000009   0x008FD6   4

```

48.73.239.44	48.73.239.44	1993	0x80000009	0x003310	4
48.73.239.55	48.73.239.55	2000	0x80000008	0x006E2A	2
48.73.239.66	48.73.239.66	1512	0x80000005	0x003945	2

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
10.1.2.2	48.73.239.22	459	0x80000006	0x00EDC7
10.1.3.1	48.73.239.11	1998	0x80000005	0x00F7BE
10.2.4.2	48.73.239.44	1729	0x80000004	0x00C2BA
10.3.4.2	48.73.239.44	1993	0x80000007	0x004B23
10.3.5.2	48.73.239.55	2000	0x80000007	0x006CEA
10.4.6.2	48.73.239.66	1511	0x80000004	0x002215

R1#show ip route ospf

10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks

```

0      10.4.6.0/30 [110/3] via 10.1.3.2, 01:37:27, GigabitEthernet2/0
                        [110/3] via 10.1.2.2, 01:46:27, GigabitEthernet1/0
0      10.2.2.2/32 [110/2] via 10.1.2.2, 01:46:27, GigabitEthernet1/0
0      10.3.3.3/32 [110/2] via 10.1.3.2, 01:37:27, GigabitEthernet2/0
0      10.6.6.6/32 [110/4] via 10.1.3.2, 01:37:27, GigabitEthernet2/0
                        [110/4] via 10.1.2.2, 01:46:27, GigabitEthernet1/0
0      10.3.5.0/30 [110/2] via 10.1.3.2, 01:37:27, GigabitEthernet2/0
0      10.2.4.0/30 [110/2] via 10.1.2.2, 01:46:27, GigabitEthernet1/0
0      10.3.4.0/30 [110/2] via 10.1.3.2, 01:37:27, GigabitEthernet2/0
0      10.4.4.4/32 [110/3] via 10.1.3.2, 01:37:27, GigabitEthernet2/0
                        [110/3] via 10.1.2.2, 01:46:27, GigabitEthernet1/0
0      10.5.5.5/32 [110/3] via 10.1.3.2, 01:37:27, GigabitEthernet2/0

```

R1#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
48.73.239.33	1	FULL/BDR	00:00:37	10.1.3.2	GigabitEthernet2/0
48.73.239.22	1	FULL/DR	00:00:35	10.1.2.2	GigabitEthernet1/0

R1#sh ip ospf interface brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Lo0	1	0	10.1.1.1/32	1	LOOP	0/0	
Gi2/0	1	0	10.1.3.1/30	1	DR	1/1	
Gi1/0	1	0	10.1.2.1/30	1	BDR	1/1	

AS 17390 - IBM

! From Router 7 - ping Router 8 Loopback 0 -----
R7#ping 10.8.8.8

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.8.8.8, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 12/21/36 ms

OSPF configuration analysis example Router 7

! ROUTER 7 -----
R7#show ip ospf database

OSPF Router with ID (130.41.46.77) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
130.41.46.77	130.41.46.77	1385	0x80000005	0x007902	2
130.41.46.88	130.41.46.88	1347	0x80000005	0x00FD63	2

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
10.7.8.2	130.41.46.88	1347	0x80000004	0x00197A

R7#show ip route ospf

10.0.0.0/8 is variably subnetted, 3 subnets, 2 masks

```

0      10.8.8.8/32 [110/2] via 10.7.8.2, 02:03:15, GigabitEthernet1/0

```

```
R7#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
130.41.46.88	1	FULL/DR	00:00:36	10.7.8.2	GigabitEthernet1/0

```
R7#show ip ospf interface brief
```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Lo0	1	0	10.7.7.7/32	1	LOOP	0/0	
Gi1/0	1	0	10.7.8.1/30	1	BDR	1/1	

AS 5511 - FTRSI (Orange - Worldwide IP Backbone)

```
From Router 10 - ping Routers 11 and 12 Loopback 0 -----  
R10#ping 10.11.11.11
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.11.11.11, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 8/20/44 ms

```
R10#ping 10.12.12.12
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 10.12.12.12, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 16/32/80 ms

OSPF configuration analysis

```
! ROUTER 10 -----  
R10#show ip ospf database
```

OSPF Router with ID (10.10.10.10) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
10.10.10.10	10.10.10.10	321	0x8000000B	0x00421C	2
10.11.11.11	10.11.11.11	401	0x8000000B	0x004113	2

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
10.10.11.1	10.10.10.10	321	0x80000005	0x004C51

Summary Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
10.10.12.0	10.10.10.10	1090	0x80000006	0x00CA27
10.10.12.0	10.11.11.11	401	0x80000003	0x00C52B
10.11.12.0	10.10.10.10	582	0x80000003	0x00CE24
10.11.12.0	10.11.11.11	401	0x80000007	0x00A745
10.12.12.12	10.10.10.10	582	0x80000003	0x005C86
10.12.12.12	10.11.11.11	401	0x80000003	0x004798
46.87.162.0	10.10.10.10	582	0x80000003	0x00ECFE
46.87.162.0	10.11.11.11	403	0x80000003	0x00D711
46.87.162.112	10.10.10.10	584	0x80000003	0x009ADD
46.87.162.112	10.11.11.11	403	0x80000003	0x0085EF

Router Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
10.10.10.10	10.10.10.10	584	0x80000009	0x00A0F8	1
10.11.11.11	10.11.11.11	404	0x80000009	0x008808	1
10.12.12.12	10.12.12.12	528	0x8000000D	0x006CA7	5

Net Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
10.10.12.1	10.10.10.10	584	0x80000003	0x008A13
10.11.12.2	10.12.12.12	528	0x80000005	0x00553B

Summary Net Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
0.0.0.0	10.10.10.10	1091	0x80000006	0x007A96
0.0.0.0	10.11.11.11	404	0x80000007	0x0063A9
10.10.10.10	10.10.10.10	1092	0x80000006	0x00AC3C
10.10.10.10	10.11.11.11	406	0x80000003	0x00A740
10.10.11.0	10.10.10.10	1094	0x80000006	0x00F301
10.10.11.0	10.11.11.11	406	0x80000008	0x00DA15
10.11.11.11	10.10.10.10	325	0x8000000C	0x008955
10.11.11.11	10.11.11.11	406	0x80000008	0x00726E

R10#sh ip route ospf

```

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
0       10.11.11.11/32 [110/2] via 10.10.11.2, 01:13:38, GigabitEthernet1/0
0       10.12.12.12/32 [110/2] via 10.10.12.2, 01:16:04, GigabitEthernet3/0
0       10.11.12.0/30 [110/2] via 10.10.12.2, 01:16:04, GigabitEthernet3/0
46.0.0.0/8 is variably subnetted, 11 subnets, 3 masks
0       46.87.162.112/32 [110/2] via 10.10.12.2, 01:16:04, GigabitEthernet3/0
0       46.87.162.0/30 [110/2] via 10.10.12.2, 01:16:04, GigabitEthernet3/0

```

R10#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.11.11.11	1	FULL/BDR	00:00:33	10.10.11.2	GigabitEthernet1/0
10.12.12.12	1	FULL/BDR	00:00:36	10.10.12.2	GigabitEthernet3/0

R10#show ip ospf interface brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Lo0	1	0	10.10.10.10/32	1	LOOP	0/0	
Gi1/0	1	0	10.10.11.1/30	1	DR	1/1	
Gi3/0	1	1	10.10.12.1/30	1	DR	1/1	

! ROUTER 11 -----

R11#show ip ospf database

OSPF Router with ID (10.11.11.11) (Process ID 1)

Router Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
10.10.10.10	10.10.10.10	461	0x8000000B	0x00421C	2
10.11.11.11	10.11.11.11	539	0x8000000B	0x004113	2

Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
10.10.11.1	10.10.10.10	461	0x80000005	0x004C51

Summary Net Link States (Area 0)

Link ID	ADV Router	Age	Seq#	Checksum
10.10.12.0	10.10.10.10	1230	0x80000006	0x00CA27
10.10.12.0	10.11.11.11	539	0x80000003	0x00C52B
10.11.12.0	10.10.10.10	722	0x80000003	0x00CE24
10.11.12.0	10.11.11.11	539	0x80000007	0x00A745
10.12.12.12	10.10.10.10	722	0x80000003	0x005C86
10.12.12.12	10.11.11.11	539	0x80000003	0x004798
46.87.162.0	10.10.10.10	722	0x80000003	0x00ECFE
46.87.162.0	10.11.11.11	541	0x80000003	0x00D711
46.87.162.112	10.10.10.10	723	0x80000003	0x009ADD
46.87.162.112	10.11.11.11	541	0x80000003	0x0085EF

Router Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
10.10.10.10	10.10.10.10	724	0x80000009	0x00A0F8	1
10.11.11.11	10.11.11.11	541	0x80000009	0x008808	1
10.12.12.12	10.12.12.12	667	0x8000000D	0x006CA7	5

Net Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
10.10.12.1	10.10.10.10	724	0x80000003	0x008A13
10.11.12.2	10.12.12.12	667	0x80000005	0x00553B

Summary Net Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
0.0.0.0	10.10.10.10	1232	0x80000006	0x007A96
0.0.0.0	10.11.11.11	541	0x80000007	0x0063A9
10.10.10.10	10.10.10.10	1232	0x80000006	0x00AC3C
10.10.10.10	10.11.11.11	541	0x80000003	0x00A740
10.10.11.0	10.10.10.10	1233	0x80000006	0x00F301
10.10.11.0	10.11.11.11	541	0x80000008	0x00DA15
10.11.11.11	10.10.10.10	464	0x8000000C	0x008955
10.11.11.11	10.11.11.11	541	0x80000008	0x00726E

R11#show ip route ospf

```

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
0    10.10.10.10/32 [110/2] via 10.10.11.1, 01:16:03, GigabitEthernet1/0
0    10.12.12.12/32 [110/2] via 10.11.12.2, 01:16:03, GigabitEthernet2/0
0    10.10.12.0/30 [110/2] via 10.11.12.2, 01:16:03, GigabitEthernet2/0
46.0.0.0/8 is variably subnetted, 11 subnets, 3 masks
0    46.87.162.112/32 [110/2] via 10.11.12.2, 01:16:03, GigabitEthernet2/0
0    46.87.162.0/30 [110/2] via 10.11.12.2, 01:16:03, GigabitEthernet2/0

```

R11#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.10.10.10	1	FULL/DR	00:00:36	10.10.11.1	GigabitEthernet1/0
10.12.12.12	1	FULL/DR	00:00:38	10.11.12.2	GigabitEthernet2/0

R11#show ip ospf interface brief

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Lo0	1	0	10.11.11.11/32	1	LOOP	0/0	
Gi1/0	1	0	10.10.11.2/30	1	BDR	1/1	
Gi2/0	1	1	10.11.12.1/30	1	BDR	1/1	

! ROUTER 12 -----
R12#show ip ospf database

OSPF Router with ID (10.12.12.12) (Process ID 1)

Router Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum	Link count
10.10.10.10	10.10.10.10	943	0x80000009	0x00A0F8	1
10.11.11.11	10.11.11.11	761	0x80000009	0x008808	1
10.12.12.12	10.12.12.12	886	0x8000000D	0x006CA7	5

Net Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
10.10.12.1	10.10.10.10	943	0x80000003	0x008A13
10.11.12.2	10.12.12.12	886	0x80000005	0x00553B

Summary Net Link States (Area 1)

Link ID	ADV Router	Age	Seq#	Checksum
0.0.0.0	10.10.10.10	1451	0x80000006	0x007A96
0.0.0.0	10.11.11.11	761	0x80000007	0x0063A9
10.10.10.10	10.10.10.10	1451	0x80000006	0x00AC3C
10.10.10.10	10.11.11.11	761	0x80000003	0x00A740
10.10.11.0	10.10.10.10	1451	0x80000006	0x00F301
10.10.11.0	10.11.11.11	763	0x80000008	0x00DA15
10.11.11.11	10.10.10.10	683	0x8000000C	0x008955
10.11.11.11	10.11.11.11	763	0x80000008	0x00726E

R12#sh ip route ospf

```

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
0 IA  10.10.10.10/32 [110/2] via 10.10.12.1, 01:21:57, GigabitEthernet3/0
0 IA  10.11.11.11/32 [110/2] via 10.11.12.1, 01:19:31, GigabitEthernet2/0
0 IA  10.10.11.0/30 [110/2] via 10.11.12.1, 01:19:31, GigabitEthernet2/0
      [110/2] via 10.10.12.1, 01:21:57, GigabitEthernet3/0
0*IA 0.0.0.0/0 [110/2] via 10.11.12.1, 01:19:31, GigabitEthernet2/0
      [110/2] via 10.10.12.1, 01:21:57, GigabitEthernet3/0

```

R12#show ip ospf neighbor

Neighbor ID	Pri	State	Dead Time	Address	Interface
10.10.10.10	1	FULL/DR	00:00:33	10.10.12.1	GigabitEthernet3/0
10.11.11.11	1	FULL/BDR	00:00:37	10.11.12.1	GigabitEthernet2/0


```
R12#show ip ospf interface brief
```

Interface	PID	Area	IP Address/Mask	Cost	State	Nbrs	F/C
Lo0	1	1	10.12.12.12/32	1	LOOP	0/0	
Lo1	1	1	46.87.162.112/32	1	LOOP	0/0	
Fa0/0	1	1	46.87.162.2/30	1	DR	0/0	
Gi3/0	1	1	10.10.12.2/30	1	BDR	1/1	
Gi2/0	1	1	10.11.12.2/30	1	DR	1/1	

Test the interfaces connectivity between the ASes

Example from AS 1273 to AS 17390

From	To	Interface	IP Address	Connectivity Success
R1	R8	Lo1	130.41.46.88	No
R1	R8	g4/0	48.73.240.2	No
R1	R8	g5/0	130.41.46.2	No
R1	R7	Lo1	130.41.46.77	No
R1	R7	f0/0	130.41.46.9	No
R1	R7	g4/0	48.73.240.6	No
R1	R7	g5/0	130.41.46.2	No

We have no connectivity between ASes, even in direct connected links. To have connectivity between ASes we need to configure BGP.

1.4 Practical Questions

1.4.1 - Create a comprehensive table presenting all the connectivity tests carried out and the respective outcome (e.g., success, failure). You don't need to provide exhaustive snapshots for all test results. Choose only three example cases, from different ASes, to include in your report and briefly comment on each selected case.

Inside ASes we have connectivity between all routers but no connectivity outside as shown previously in the Test and Validation part and in the previous question.

1.4.2 - Identification of the ASes entities involved in the lab topology

ASN	AS Name	Country
701	Verizon Business	US
64513	(Private AS)	
17390	IBM	US
4637	Telstra Global (APNIC)	Hong Kong
1273	Vodafone	UK
1	Level 3 Parent, LLC (Historical ARPANET)	US
20717	DE-CIX Management GmbH	Germany
5511	FTRSI (Orange - Worldwide IP Backbone)	France
23344	Disney Worldwide Services, Inc.	US

1.4.3 - Explain the concept of an Autonomous System (AS) in the Internet architecture and provide examples associated with the Portuguese Internet ecosystem

In the Internet architecture, an Autonomous System (AS) is a large network or group of networks controlled by a single

organization that operates under a unified routing policy. Each AS is assigned a unique Autonomous System Number (ASN) for identification when exchanging traffic with other networks.

Portugal's Internet infrastructure consists of numerous independent networks. There 123 ASNs allocated to Portugal and are managed by various Portuguese ISPs, telecommunications companies, universities, government agencies, banks, or other large enterprises.

List of Portuguese ASes

1.4.4 - Classification of each AS in the lab project as Tier-1 or Tier-2. For each classification, describe the evidence you find in the lab topology that justifies it

Classification of Autonomous Systems

AS Number	AS Name / Entity	Classification	Evidence from Lab Topology
AS701	Verizon Business	Tier-1	Described as a "Tier 1" AS in the project overview (Section 3.2). It peers with multiple other major ASes (AS1273, AS17390) and does not need to purchase transit from anyone, fitting the definition of a top-level provider.
AS1273	Vodafone	Tier-2	A large telecommunications provider. While global, it acts as a customer of AS701 (purchasing transit), which is a key indicator of a Tier-2. It also peers with multiple other ASes (AS17390, AS20717, AS5511) to reduce its transit costs.
AS4637	Telstra Global	Tier-2	A dominant national provider in Australia, but on the global stage, it functions as a Tier-2. The topology shows it connecting to other major ASes (AS1, AS20717, AS23344), engaging in a mix of peering and transit relationships.
AS5511	Orange (FTRSI)	Tier-2	Another major international telecom operator. Its role in the lab is that of a large provider that peers with others (e.g., via AS20717) but is not depicted as part of the exclusive Tier-1 core.
AS17390	IBM	Enterprise (Tier-2)	This is a corporate enterprise network. Its purpose is to connect its own services (like IBM Cloud) to the internet, not to sell transit to others. It purchases transit from AS1273, making it a classic Tier-3 / "stub" AS.
AS23344	Disney	Enterprise (Tier-2)	This is a content/enterprise network for Disney. It has only one eBGP peering to the internet (with AS4637) and is configured to receive only a default route, which is typical behavior for an enterprise customer.
AS20717	DE-CIX Management GmbH	Internet Exchange (IXP)	This is not a transit provider. It is the AS for the DE-CIX Internet Exchange Point. Its function is to provide a neutral fabric for other ASes to peer, which places it outside the Tier hierarchy.
AS64513	(Private AS)	Unclassifiable	This is a Private AS number (range 64512-65534) used internally, likely for a multihomed customer within AS17390. It is not advertised to the global internet and has no tier classification.

Justification and Analysis

The classification is based on the core differentiator between tiers: **transit and peering relationships**.

- Tier-1 Evidence:** AS701 and AS1 are presented as the foundational providers in the topology. A key piece of evidence is that **AS1273 (Vodafone) purchases transit from AS701**. The fact that one AS is a customer of another immediately defines the provider (AS701) as higher in the hierarchy. Tier-1 networks form a club that can reach the entire internet via settlement-free peering with each other, and they sell transit to all others.
- Tier-2 Evidence:** AS1273, AS4637, and AS5511 are all massive telecommunications companies. However, their behavior in the lab aligns with Tier-2s:

- They operate large backbones and provide transit to their own customers (e.g., AS1273 provides transit to AS17390).
 - Crucially, they also **purchase transit** from Tier-1s (as seen with AS1273 and AS701).
 - They engage in extensive **peering** at neutral points (like AS20717) to exchange traffic directly with other networks, reducing their reliance on expensive Tier-1 transit links. This mixed strategy of buying transit and peering for free is the hallmark of a Tier-2.
3. **Special Case - IXP:** AS20717 (DE-CIX) is not a tiered AS. It is infrastructure. It does not buy or sell transit. It exists so that ASes from all tiers (including the Tier-1s, Tier-2s, and Enterprises in this lab) can connect and exchange traffic efficiently.

1.4.5 - Table showcasing all the peering relations established in the provided topology

AS	Peers
1273	20717, 4637, 701, 17390, 5511
17390	701, 1273
701	17390, 1273, 4637
4637	701, 1273, 1, 5511
1	4637
20717	1273, 5511
5511	20717, 23344, 1273, 4537
23344	5511

1.4.6 - Explain how a Tier-2 benefits from peering instead of buying everything from a Tier-1

A Tier-2 ISP operates on a simple principle: it must pay a Tier-1 for access to the entire internet. However, a significant portion of its traffic is often destined for a handful of other large networks (like other Tier-2s, cloud providers, or content giants like Google and Netflix).

Peering is the strategy of connecting directly with these specific networks to exchange traffic, creating a "shortcut" that bypasses the Tier-1 transit path.

Key Benefits of Peering for a Tier-2:

- Cost Reduction
- Improved Performance and Lower Latency
- Increased Reliability and Redundancy
- Greater Control over Routing

1.4.7 - Identify the neutral public peering interconnections in this lab topology. Elaborate on why they are called neutral and provide examples of real-world implementations of such public interconnections

The neutral public peering interconnection is represented by AS20717 - DE-CIX Management GmbH.

Instead of setting up direct, individual links between each other (private peering), these ASes can all connect to the shared switch fabric of AS20717

Examples of real world implementations:

- DE-CIX (Frankfurt) - de-cix.net (lab)
- AMS-IX (Amsterdam) - ams-ix.net
- LINX (London) - linx.net
- EQUINIX IX (Various) - equinix.com/ix
- NYIIX (New York) - nyiix.net

By connecting to these neutral hubs, the Tier-2 ISPs can exchange a large volume of their traffic directly, locally, and for free, achieving the significant benefits of peering.

1.4.8 - Explain the role of R12 in AS 5511, and how are its interfaces divided between the OSPF areas involved

R12 is Router inside a Stub Area

R10 and R11 serves as an Area Border Router (ABR). This is its primary and most important role.

As an ABR, R10 and R11 has the following key responsibilities:

Connecting Different OSPF Areas: It has interfaces in two separate OSPF areas: the backbone area (Area 0) and a stub area (Area 1). It is the sole router providing a pathway between these two parts of the network.

Summarizing and Filtering LSAs: It controls the flow of Link-State Advertisements (LSAs) between Area 0 and Area 1.

Injecting a Default Route: As Area 1 is a stub area, R10 and R11 does not flood external LSAs (Type 5 LSAs) into it. Instead, it injects a default route (0.0.0.0/0) into Area 1, telling all routers in that area, "if you don't have a more specific route, send the traffic to me to reach external destinations."

1.4.9 - Explain what a stub area is and discuss the resulting advantages and potential limitations. In your discussion, please detail under what conditions would multi-area OSPF be preferred over a single backbone area in real networks

In the OSPF (Open Shortest Path First) routing protocol, the network is divided into **areas** to create a hierarchical structure. This hierarchy is crucial for controlling the flow of routing information and improving scalability.

A **stub area** is a specific type of OSPF area designed to minimize the size of the routing table and the amount of Link-State Advertisement (LSA) traffic flooding into it.

Core Concept: A stub area is configured to block external routes from entering. These external routes are routes redistributed into OSPF from other routing protocols (like BGP, EIGRP, or static routes). They are carried by **Type 5 LSAs**.

Instead of learning these potentially vast numbers of external routes, a router in a stub area uses a **default route (0.0.0.0/0)**, injected by the Area Border Router (ABR), to reach any destination outside the OSPF domain.

Key Characteristics of a Stub Area:

- **Blocks Type 5 LSAs:** The ABR does not flood AS External LSAs (Type 5) into the stub area.
- **Uses a Default Route:** The ABR automatically generates and advertises a default route (Type 3 LSA) into the stub area.
- **Cannot be a Transit Area:** Traffic from one non-backbone area cannot use a stub area to reach another non-backbone area. Virtual Links cannot be configured through a stub area.
- **No ASBRs Allowed:** An Autonomous System Boundary Router (ASBR) cannot be located inside a stub area, as its job is to generate the very Type 5 LSAs that are blocked.

Advantages

1. **Reduced Routing Table Size:** This is the primary benefit. By replacing thousands of external routes with a single default route, the routing tables of routers within the stub area are significantly smaller. This saves memory (RAM).
2. **Reduced LSA Traffic and CPU Overload:** By blocking Type 5 LSAs, there is less Link-State Update traffic flooding within the area. This reduces the CPU cycles required for OSPF SPF (Shortest Path First) calculations, especially when external networks are flapping.
3. **Improved Stability:** Instabilities in external networks (e.g., Internet route changes) are completely isolated from the routers inside the stub area. This makes the internal OSPF domain more stable.
4. **Simpler Configuration and Management:** For remote branch offices that only need a default path to the central data center/Internet, a stub area configuration is simple and logical.

Potential Limitations

1. **Suboptimal Routing:** This is the most significant trade-off. Because internal routers only have a default route, they have no topological awareness of external paths. All traffic destined outside the OSPF domain is sent to the ABR, which may not be the most efficient path if multiple exits exist.
2. **Loss of Granular Control:** Network administrators lose the ability to make precise routing decisions for external destinations within the stub area, as the specific routes are unknown.
3. **Restrictive Topology:** The requirement that no ASBR can reside in a stub area limits network design flexibility. You cannot have a branch office that also connects to, for example, a partner network via BGP if it's configured as a stub area (you would use an NSSA instead).

1.4.10 - Discuss why the subnet 46.87.162.0/24 was not placed on the backbone area, considering the OSPF design

principles

The fundamental rule is: "Keep the backbone clean." Area 0 should be a high-speed transit core for inter-area traffic, not a place for end-user or server subnets.

Key Reasons for Stub Area Placement:

- Hierarchy & Stability
- Reduced Routing Overhead
- Controlled Traffic Flow
- Scalability

Putting end-user subnets in Area 0 violates OSPF hierarchy, unnecessarily exposes the core to edge network instability, and eliminates the benefits of LSA filtering, leading to a less scalable and resilient network.

2 - Phase 2 - iBGP and eBGP Without Routing Policies

2.1 - Objectives

In this phase we have as objectives:

- Establish eBGP sessions between the AS's
- Inside the AS 1273, establish iBGP sessions between the clients and the two route reflectors, R3 and R4 to avoid the full mesh
- Server subnet public IPs are listed in the internet routing table
- Implement connectivity between the ASes from any routers using the Lo1 and from any server

2.2 - Implementation

BGP Configuration

AS 1273 - Vodafone , R1 and R4 as examples

```
! Router 1 =====
!
! =====
!                               STATIC ROUTES
! =====
!
ip route 48.73.239.0 255.255.255.0 Null0
ip route 48.73.240.0 255.255.252.0 Null0
ip route 63.25.64.0 255.255.192.0 Null0
!
! =====
!                               BGP
! =====
!
router bgp 1273
  bgp router-id 10.1.1.1
  no synchronization
  bgp log-neighbor-changes
  !
  ! iBGP - adjacencies in Loopback interfaces -----
  neighbor iBGP peer-group
  neighbor iBGP remote-as 1273
  neighbor iBGP update-source Loopback0
  neighbor iBGP next-hop-self
  neighbor iBGP soft-reconfiguration inbound
  !
  neighbor 10.3.3.3 peer-group iBGP
  neighbor 10.4.4.4 peer-group iBGP
  !
  ! eBGP - adjacencies in physical interfaces -----
  neighbor 48.73.240.2 remote-as 17390
  neighbor 48.73.240.2 soft-reconfiguration inbound
  !
  ! Advertise Routes -----
  network 48.73.239.11 mask 255.255.255.255      ! Loopback 1
  network 48.73.239.0 mask 255.255.255.0
  network 48.73.240.0 mask 255.255.252.0
  network 63.25.64.0 mask 255.255.192.0
  no auto-summary                               ! Do not summarize routes
```

```
! Router 4 =====
!
! =====
!                               STATIC ROUTES
! =====
!
ip route 48.73.239.0 255.255.255.0 Null0
ip route 48.73.240.0 255.255.252.0 Null0
ip route 63.25.64.0 255.255.192.0 Null0
```

```

!
! =====
!                                     BGP
! =====
!
router bgp 1273
  bgp router-id 10.4.4.4
  bgp cluster-id 1273
  no synchronization
  bgp log-neighbor-changes
  ! iBGP -----
  neighbor iBGP peer-group
  neighbor iBGP remote-as 1273
  neighbor iBGP update-source Loopback0
  neighbor iBGP route-reflector-client
  neighbor iBGP next-hop-self
  neighbor iBGP soft-reconfiguration inbound
  !
  neighbor iBGP_RR peer-group
  neighbor iBGP_RR remote-as 1273
  neighbor iBGP_RR update-source Loopback0
  neighbor iBGP_RR soft-reconfiguration inbound
  !
  neighbor 10.1.1.1 peer-group iBGP
  neighbor 10.2.2.2 peer-group iBGP
  neighbor 10.3.3.3 peer-group iBGP_RR
  neighbor 10.5.5.5 peer-group iBGP
  neighbor 10.6.6.6 peer-group iBGP
  !
  ! Advertise Routes -----
  network 48.73.239.44 mask 255.255.255.255    ! Lo1
  network 48.73.239.0 mask 255.255.255.0
  network 48.73.240.0 mask 255.255.252.0
  network 63.25.64.0 mask 255.255.192.0
  no auto-summary

```

AS 17390 - IBM, Router 7 as example

```

! Router 7 =====
!
! =====
!                                     STATIC ROUTES
! =====
!
ip route 130.41.46.0 255.255.255.0 Null0
!
! =====
!                                     BGP
! =====
!
router bgp 17390
  bgp router-id 10.7.7.7
  no synchronization
  bgp log-neighbor-changes
  !
  ! iBGP -----
  neighbor 10.8.8.8 remote-as 17390
  neighbor 10.8.8.8 update-source Loopback0
  neighbor 10.8.8.8 next-hop-self
  neighbor 10.8.8.8 soft-reconfiguration inbound
  !
  ! eBGP -----
  neighbor 64.112.0.5 remote-as 701
  neighbor 64.112.0.5 soft-reconfiguration inbound
  neighbor 130.41.46.10 remote-as 64513
  neighbor 130.41.46.10 soft-reconfiguration inbound
  neighbor 48.73.240.5 remote-as 1273
  neighbor 48.73.240.5 soft-reconfiguration inbound
  !
  ! Advertise Routes -----
  network 130.41.46.77 mask 255.255.255.255    ! Lo1
  network 130.41.46.0 mask 255.255.255.0
  no auto-summary

```


AS 64513 - Private (inside IBM)

```
! Router 9 =====
!  
! =====  
!                               STATIC ROUTES  
! =====  
!  
ip route 130.41.47.0 255.255.255.0 Null0  
!  
! =====  
!                               BGP  
! =====  
!  
router bgp 64513  
  bgp router-id 10.9.9.9  
  no synchronization  
  bgp log-neighbor-changes  
  !  
  ! eBGP -----  
  neighbor 130.41.46.9 remote-as 17390  
  neighbor 130.41.46.9 soft-reconfiguration inbound  
  neighbor 130.41.46.5 remote-as 17390  
  neighbor 130.41.46.5 soft-reconfiguration inbound  
  !  
  ! Advertise Routes -----  
  network 130.41.47.99 mask 255.255.255.255    ! Lo1  
  network 130.41.47.0 mask 255.255.255.0  
  no auto-summary
```

AS 5511 - Orange

```
! Router 10 =====
!  
! =====  
!                               STATIC ROUTES  
! =====  
!  
ip route 46.87.162.0 255.255.255.0 Null0  
ip route 46.88.20.0 255.255.255.0 Null0  
ip route 46.88.22.0 255.255.255.0 Null0  
ip route 46.89.36.0 255.255.255.0 Null0  
ip route 46.89.38.0 255.255.255.0 Null0  
ip route 46.92.2.0 255.255.255.0 Null0  
!  
! =====  
!                               BGP  
! =====  
!  
!  
router bgp 5511  
  bgp router-id 10.10.10.10  
  no synchronization  
  bgp log-neighbor-changes  
  !  
  ! iBGP -----  
  neighbor 10.11.11.11 remote-as 5511  
  neighbor 10.11.11.11 update-source Loopback0  
  neighbor 10.11.11.11 next-hop-self  
  neighbor 10.11.11.11 soft-reconfiguration inbound  
  neighbor 10.12.12.12 remote-as 5511  
  neighbor 10.12.12.12 update-source Loopback0  
  neighbor 10.12.12.12 next-hop-self  
  neighbor 10.12.12.12 soft-reconfiguration inbound  
  !  
  ! eBGP -----  
  neighbor 46.88.20.2 remote-as 20717  
  neighbor 46.88.20.2 soft-reconfiguration inbound  
  neighbor 48.73.240.17 remote-as 1273  
  neighbor 48.73.240.17 soft-reconfiguration inbound  
  neighbor 211.176.129.1 remote-as 4637  
  neighbor 211.176.129.1 soft-reconfiguration inbound  
  !
```

```

! Advertise Routes -----
network 46.87.162.110 mask 255.255.255.255      ! Lo1
network 46.87.162.0 mask 255.255.255.0
network 46.88.20.0 mask 255.255.255.0
network 46.88.22.0 mask 255.255.255.0
network 46.89.36.0 mask 255.255.255.0
network 46.89.38.0 mask 255.255.255.0
network 46.92.2.0 mask 255.255.255.0
no auto-summary

```

```

! Router 11 =====
!
! =====
!                               STATIC ROUTES
! =====
!
ip route 46.87.162.0 255.255.255.0 Null0
ip route 46.88.20.0 255.255.255.0 Null0
ip route 46.88.22.0 255.255.255.0 Null0
ip route 46.89.36.0 255.255.255.0 Null0
ip route 46.89.38.0 255.255.255.0 Null0
ip route 46.92.2.0 255.255.255.0 Null0
!
! =====
!                               BGP
! =====
!
router bgp 5511
  bgp router-id 10.11.11.11
  no synchronization
  bgp log-neighbor-changes
  !
  ! iBGP -----
  neighbor 10.10.10.10 remote-as 5511
  neighbor 10.10.10.10 update-source Loopback0
  neighbor 10.10.10.10 next-hop-self
  neighbor 10.10.10.10 soft-reconfiguration inbound
  neighbor 10.12.12.12 remote-as 5511
  neighbor 10.12.12.12 update-source Loopback0
  neighbor 10.12.12.12 next-hop-self
  neighbor 10.12.12.12 soft-reconfiguration inbound
  !
  ! eBGP -----
  neighbor 46.88.20.6 remote-as 23344
  neighbor 46.88.20.6 soft-reconfiguration inbound
  !
  ! Advertise Routes -----
  network 46.87.162.111 mask 255.255.255.255      ! Lo1
  network 46.87.162.0 mask 255.255.255.0
  network 46.88.20.0 mask 255.255.255.0
  network 46.88.22.0 mask 255.255.255.0
  network 46.89.36.0 mask 255.255.255.0
  network 46.89.38.0 mask 255.255.255.0
  network 46.92.2.0 mask 255.255.255.0
  no auto-summary

```

```

! Router 12 =====
!
! =====
!                               BGP
! =====
!
router bgp 5511
  bgp router-id 10.12.12.12
  no synchronization
  bgp log-neighbor-changes
  !
  ! iBGP -----
  neighbor 10.10.10.10 remote-as 5511
  neighbor 10.10.10.10 update-source Loopback0
  neighbor 10.10.10.10 next-hop-self
  neighbor 10.10.10.10 soft-reconfiguration inbound
  neighbor 10.11.11.11 remote-as 5511

```

```

neighbor 10.11.11.11 update-source Loopback0
neighbor 10.11.11.11 next-hop-self
neighbor 10.11.11.11 soft-reconfiguration inbound
!
! Advertise Routes -----
network 46.87.162.112 mask 255.255.255.255    ! Lo1
network 46.87.162.1 mask 255.255.255.255      ! Server5

```

AS 23344 - Disney

```

! Router 13 =====
!
! =====
!                               STATIC ROUTES
! =====
!
ip route 158.23.228.0 255.255.255.0 Null0
!
! =====
!                               BGP
! =====
!
router bgp 23344
  bgp router-id 10.13.13.13
  no synchronization
  bgp log-neighbor-changes
  !
  ! eBGP -----
  neighbor 46.88.20.5 remote-as 5511
  neighbor 46.88.20.5 soft-reconfiguration inbound
  !
  ! Advertise Routes -----
  network 158.23.228.113 mask 255.255.255.255    ! Lo1
  network 158.23.228.0 mask 255.255.255.0
  network 158.23.228.0 mask 255.255.255.252      ! avoid black-hole / access server6
  no auto-summary

```

AS 701 - Verizon

```

! Router 14 =====
!
! =====
!                               STATIC ROUTES
! =====
!
ip route 64.96.0.0 255.240.0.0 Null0
ip route 64.112.0.0 255.240.0.0 Null0
ip route 65.0.204.0 255.255.255.0 Null0
ip route 64.96.0.0 255.255.255.252 fastEthernet0/0
!
! =====
!                               BGP
! =====
!
router bgp 701
  bgp router-id 10.14.14.14
  no synchronization
  bgp log-neighbor-changes
  !
  ! eBGP -----
  neighbor 64.112.0.10 remote-as 4637
  neighbor 64.112.0.10 soft-reconfiguration inbound
  neighbor 64.112.0.2 remote-as 1273
  neighbor 64.112.0.2 soft-reconfiguration inbound
  neighbor 64.112.0.6 remote-as 17390
  neighbor 64.112.0.6 soft-reconfiguration inbound
  !
  ! Advertise Routes -----
  network 64.96.0.114 mask 255.255.255.255    ! Lo1
  network 64.96.0.0 mask 255.240.0.0
  network 64.96.0.0 mask 255.255.255.252      ! avoid black-hole / access server3
  network 64.112.0.0 mask 255.240.0.0
  network 65.0.204.0 mask 255.255.255.0

```

```
no auto-summary
```

AS 4637 - Telstra

```
! Router 15 =====
!
! =====
!                               STATIC ROUTES
! =====
!
ip route 211.176.128.0 255.255.255.0 Null0
ip route 211.176.129.0 255.255.255.0 Null0
ip route 211.176.130.0 255.255.254.0 Null0
ip route 211.176.132.0 255.255.255.0 Null0
ip route 211.176.135.0 255.255.255.0 Null0
ip route 211.176.136.0 255.255.252.0 Null0
ip route 211.176.137.0 255.255.255.0 Null0
ip route 211.176.138.0 255.255.255.0 Null0
ip route 211.176.139.0 255.255.255.0 Null0
ip route 211.176.140.0 255.255.255.0 Null0
ip route 211.176.141.0 255.255.255.0 Null0
ip route 211.176.142.0 255.255.255.0 Null0
!
! =====
!                               BGP
! =====
!
router bgp 4637
  bgp router-id 10.15.15.15
  no synchronization
  bgp log-neighbor-changes
  !
  ! eBGP -----
  neighbor 64.112.0.9 remote-as 701
  neighbor 64.112.0.9 soft-reconfiguration inbound
  neighbor 48.73.240.13 remote-as 1273
  neighbor 48.73.240.13 soft-reconfiguration inbound
  neighbor 211.176.129.2 remote-as 5511
  neighbor 211.176.129.2 soft-reconfiguration inbound
  neighbor 211.176.129.6 remote-as 1
  neighbor 211.176.129.6 soft-reconfiguration inbound
  !
  ! Advertise Routes -----
  network 211.176.128.115 mask 255.255.255.255    ! Lo1
  network 211.176.128.0 mask 255.255.255.0
  network 211.176.128.0 mask 255.255.255.252    ! avoid black-hole / access server4
  network 211.176.129.0 mask 255.255.255.0
  network 211.176.130.0 mask 255.255.254.0
  network 211.176.132.0 mask 255.255.255.0
  network 211.176.135.0 mask 255.255.255.0
  network 211.176.136.0 mask 255.255.252.0
  network 211.176.137.0 mask 255.255.255.0
  network 211.176.138.0 mask 255.255.255.0
  network 211.176.139.0 mask 255.255.255.0
  network 211.176.140.0 mask 255.255.255.0
  network 211.176.141.0 mask 255.255.255.0
  network 211.176.142.0 mask 255.255.255.0
  no auto-summary
```

AS 20717 - DE-CIX

```
! Router 16 =====
! =====
!                               BGP
! =====
!
router bgp 20717
  bgp router-id 10.16.16.16
  no synchronization
  bgp log-neighbor-changes
  !
  ! eBGP -----
  neighbor 48.73.240.21 remote-as 1273
```

```
neighbor 48.73.240.21 soft-reconfiguration inbound
neighbor 46.88.20.1 remote-as 5511
neighbor 46.88.20.1 soft-reconfiguration inbound
! Without Loopback 1 to advertise
no auto-summary
```

AS 1 - Level 3 Parent

```
! BadGuy =====
! =====
!                               BGP
! =====
!
router bgp 1
  bgp router-id 10.66.66.66
  no synchronization
  bgp log-neighbor-changes
  !
  ! eBGP -----
  neighbor 211.176.129.5 remote-as 4637
  neighbor 211.176.129.5 soft-reconfiguration inbound
  no auto-summary
```

2.3 - Test and Validation

BGP Status

AS 1273 - Vodafone

```
! Router 1 =====
!
R1#show ip bgp summary
BGP router identifier 10.1.1.1, local AS number 1273
BGP table version is 698, main routing table version 698
48 network entries using 6336 bytes of memory
109 path entries using 5668 bytes of memory
13/9 BGP path/bestpath attribute entries using 2184 bytes of memory
5 BGP rrinfo entries using 120 bytes of memory
7 BGP AS-PATH entries using 168 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 3 (at peak 4) using 96 bytes of memory
BGP using 14572 total bytes of memory
BGP activity 72/24 prefixes, 454/345 paths, scan interval 60 secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
10.3.3.3	4	1273	318	261	698	0	0	03:47:15	47
10.4.4.4	4	1273	306	243	698	0	0	03:56:16	47
48.73.240.2	4	17390	282	268	698	0	0	03:56:32	11

```
! Router 2 =====
!
R2#show ip bgp summary
BGP router identifier 10.2.2.2, local AS number 1273
BGP table version is 802, main routing table version 802
48 network entries using 6336 bytes of memory
97 path entries using 5044 bytes of memory
9/8 BGP path/bestpath attribute entries using 1512 bytes of memory
5 BGP rrinfo entries using 120 bytes of memory
6 BGP AS-PATH entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 13188 total bytes of memory
BGP activity 97/49 prefixes, 429/332 paths, scan interval 60 secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
10.3.3.3	4	1273	344	265	802	0	0	03:49:31	46
10.4.4.4	4	1273	330	256	802	0	0	04:15:19	46

! Router 3 =====
!

R3#show ip bgp summary

BGP router identifier 10.3.3.3, local AS number 1273

BGP table version is 356, main routing table version 356

48 network entries using 6336 bytes of memory

88 path entries using 4576 bytes of memory

14/9 BGP path/bestpath attribute entries using 2352 bytes of memory

8 BGP AS-PATH entries using 192 bytes of memory

0 BGP route-map cache entries using 0 bytes of memory

0 BGP filter-list cache entries using 0 bytes of memory

Bitfield cache entries: current 3 (at peak 5) using 96 bytes of memory

BGP using 13552 total bytes of memory

BGP activity 86/38 prefixes, 314/226 paths, scan interval 60 secs

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
10.1.1.1	4	1273	238	281	356	0	0	03:50:48	10
10.2.2.2	4	1273	233	281	356	0	0	03:50:33	5
10.4.4.4	4	1273	276	280	356	0	0	03:51:03	4
10.5.5.5	4	1273	249	281	356	0	0	03:50:43	9
10.6.6.6	4	1273	265	281	356	0	0	03:51:02	30
48.73.240.6	4	17390	282	260	356	0	0	03:51:03	25

! Router 4 =====
!

R4#show ip bgp summary

BGP router identifier 10.4.4.4, local AS number 1273

BGP table version is 495, main routing table version 495

48 network entries using 6336 bytes of memory

69 path entries using 3588 bytes of memory

9/8 BGP path/bestpath attribute entries using 1512 bytes of memory

6 BGP AS-PATH entries using 144 bytes of memory

0 BGP route-map cache entries using 0 bytes of memory

0 BGP filter-list cache entries using 0 bytes of memory

Bitfield cache entries: current 2 (at peak 3) using 64 bytes of memory

BGP using 11644 total bytes of memory

BGP activity 96/48 prefixes, 376/307 paths, scan interval 60 secs

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
10.1.1.1	4	1273	272	348	495	0	0	04:01:11	10
10.2.2.2	4	1273	258	332	495	0	0	04:17:45	5
10.3.3.3	4	1273	343	339	495	0	0	03:52:26	11
10.5.5.5	4	1273	280	330	495	0	0	04:17:39	9
10.6.6.6	4	1273	291	332	495	0	0	04:17:54	30

! Router 5 =====
!

R5#sh ip bgp summary

BGP router identifier 10.5.5.5, local AS number 1273

BGP table version is 702, main routing table version 702

48 network entries using 6336 bytes of memory

113 path entries using 5876 bytes of memory

12/8 BGP path/bestpath attribute entries using 2016 bytes of memory

5 BGP rrinfo entries using 120 bytes of memory

9 BGP AS-PATH entries using 216 bytes of memory

0 BGP route-map cache entries using 0 bytes of memory

0 BGP filter-list cache entries using 0 bytes of memory

Bitfield cache entries: current 3 (at peak 4) using 96 bytes of memory

BGP using 14660 total bytes of memory

BGP activity 77/29 prefixes, 526/413 paths, scan interval 60 secs

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
10.3.3.3	4	1273	347	302	702	0	0	03:52:56	42
10.4.4.4	4	1273	332	282	702	0	0	04:18:30	42
64.112.0.1	4	701	214	319	702	0	0	02:45:54	25

! Router 6 =====
!

R6#show ip bgp summary

BGP router identifier 10.6.6.6, local AS number 1273

BGP table version is 571, main routing table version 571

48 network entries using 6336 bytes of memory

```

129 path entries using 6708 bytes of memory
17/9 BGP path/bestpath attribute entries using 2856 bytes of memory
5 BGP rrinfo entries using 120 bytes of memory
13 BGP AS-PATH entries using 312 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 2 (at peak 4) using 64 bytes of memory
BGP using 16396 total bytes of memory
BGP activity 77/29 prefixes, 617/488 paths, scan interval 60 secs

```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
10.3.3.3	4	1273	348	311	571	0	0	03:54:04	21
10.4.4.4	4	1273	334	292	571	0	0	04:19:34	21
48.73.240.14	4	4637	216	335	571	0	0	02:36:39	31
48.73.240.18	4	5511	314	329	571	0	0	03:10:40	26
48.73.240.22	4	20717	185	313	571	0	0	04:20:15	26

AS 17390 - IBM

```

! Router 7 =====
!
R7#show ip bgp summary
BGP router identifier 10.7.7.7, local AS number 17390
BGP table version is 521, main routing table version 521
48 network entries using 6336 bytes of memory
130 path entries using 6760 bytes of memory
20/9 BGP path/bestpath attribute entries using 3360 bytes of memory
11 BGP AS-PATH entries using 280 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 2 (at peak 4) using 64 bytes of memory
BGP using 16800 total bytes of memory
BGP activity 68/20 prefixes, 552/422 paths, scan interval 60 secs

```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
10.8.8.8	4	17390	291	314	521	0	0	04:16:39	42
48.73.240.5	4	1273	308	337	521	0	0	04:02:32	42
64.112.0.5	4	701	215	342	521	0	0	02:55:02	42
130.41.46.10	4	64513	216	322	521	0	0	04:17:23	2

```

! Router 8 =====
!
R8#show ip bgp summary
BGP router identifier 10.8.8.8, local AS number 17390
BGP table version is 489, main routing table version 489
48 network entries using 6336 bytes of memory
93 path entries using 4836 bytes of memory
16/9 BGP path/bestpath attribute entries using 2688 bytes of memory
8 BGP AS-PATH entries using 192 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 2 (at peak 4) using 64 bytes of memory
BGP using 14116 total bytes of memory
BGP activity 74/26 prefixes, 378/285 paths, scan interval 60 secs

```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
10.7.7.7	4	17390	314	292	489	0	0	04:16:59	46
48.73.240.1	4	1273	301	315	489	0	0	04:11:54	42
130.41.46.6	4	64513	217	308	489	0	0	04:17:43	2

AS 64513 - Private (inside IBM)

```

! Router 9 =====
!
R9#show ip bgp summary
BGP router identifier 10.9.9.9, local AS number 64513
BGP table version is 358, main routing table version 358
48 network entries using 6336 bytes of memory
94 path entries using 4888 bytes of memory
10/7 BGP path/bestpath attribute entries using 1680 bytes of memory
7 BGP AS-PATH entries using 184 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory

```

0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 2) using 32 bytes of memory
BGP using 13120 total bytes of memory
BGP activity 62/14 prefixes, 204/110 paths, scan interval 60 secs

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
130.41.46.5	4	17390	308	217	358	0	0	02:50:50	46
130.41.46.9	4	17390	323	217	358	0	0	02:50:50	46

AS 5511 - Orange

! Router 10 =====
!

R10#show ip bgp summary
BGP router identifier 10.10.10.10, local AS number 5511
BGP table version is 224, main routing table version 224
48 network entries using 6336 bytes of memory
112 path entries using 5824 bytes of memory
19/9 BGP path/bestpath attribute entries using 3192 bytes of memory
15 BGP AS-PATH entries using 392 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 2 (at peak 4) using 64 bytes of memory
BGP using 15808 total bytes of memory
BGP activity 57/9 prefixes, 251/139 paths, scan interval 60 secs

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
10.11.11.11	4	5511	211	239	224	0	0	03:07:51	10
10.12.12.12	4	5511	204	228	224	0	0	03:10:40	1
46.88.20.2	4	20717	133	222	224	0	0	03:20:09	22
48.73.240.17	4	1273	226	222	224	0	0	03:20:28	36
211.176.129.1	4	4637	151	231	224	0	0	02:46:48	36

! Router 11 =====
!

R11#sh ip bgp summary
BGP router identifier 10.11.11.11, local AS number 5511
BGP table version is 176, main routing table version 176
48 network entries using 6336 bytes of memory
54 path entries using 2808 bytes of memory
9/8 BGP path/bestpath attribute entries using 1512 bytes of memory
6 BGP AS-PATH entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 3 (at peak 4) using 96 bytes of memory
BGP using 10896 total bytes of memory
BGP activity 51/3 prefixes, 77/23 paths, scan interval 60 secs

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
10.10.10.10	4	5511	207	191	176	0	0	03:08:19	43
10.12.12.12	4	5511	192	191	176	0	0	03:08:44	1
46.88.20.6	4	23344	114	203	176	0	0	03:08:32	3

! Router 12 =====
!

R12#sh ip bgp summary
BGP router identifier 10.12.12.12, local AS number 5511
BGP table version is 175, main routing table version 175
48 network entries using 6336 bytes of memory
54 path entries using 2808 bytes of memory
9/8 BGP path/bestpath attribute entries using 1512 bytes of memory
6 BGP AS-PATH entries using 144 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 10832 total bytes of memory
BGP activity 51/3 prefixes, 77/23 paths, scan interval 60 secs

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
10.10.10.10	4	5511	209	193	175	0	0	03:11:33	43
10.11.11.11	4	5511	192	192	175	0	0	03:09:09	10

AS 23344 - Disney

```
! Router 13 =====
!  
R13#show ip bgp summary  
BGP router identifier 10.13.13.13, local AS number 23344  
BGP table version is 542, main routing table version 542  
48 network entries using 6336 bytes of memory  
48 path entries using 2496 bytes of memory  
9/8 BGP path/bestpath attribute entries using 1512 bytes of memory  
6 BGP AS-PATH entries using 160 bytes of memory  
0 BGP route-map cache entries using 0 bytes of memory  
0 BGP filter-list cache entries using 0 bytes of memory  
Bitfield cache entries: current 1 (at peak 2) using 32 bytes of memory  
BGP using 10536 total bytes of memory  
BGP activity 180/132 prefixes, 241/193 paths, scan interval 60 secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
46.88.20.5	4	5511	325	166	542	0	0	01:52:12	45

AS 701 - Verizon

```
! Router 14 =====
!  
R14#show ip bgp summary  
BGP router identifier 10.14.14.14, local AS number 701  
BGP table version is 94, main routing table version 94  
48 network entries using 6336 bytes of memory  
120 path entries using 6240 bytes of memory  
21/9 BGP path/bestpath attribute entries using 3528 bytes of memory  
17 BGP AS-PATH entries using 440 bytes of memory  
0 BGP route-map cache entries using 0 bytes of memory  
0 BGP filter-list cache entries using 0 bytes of memory  
Bitfield cache entries: current 1 (at peak 2) using 32 bytes of memory  
BGP using 16576 total bytes of memory  
BGP activity 65/17 prefixes, 196/76 paths, scan interval 60 secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
64.112.0.2	4	1273	190	121	94	0	0	01:49:16	43
64.112.0.6	4	17390	192	121	94	0	0	01:49:17	29
64.112.0.10	4	4637	127	130	94	0	0	01:39:21	43

AS 4637 - Telstra

```
! Router 15 =====
!  
R15#show ip bgp summary  
BGP router identifier 10.15.15.15, local AS number 4637  
BGP table version is 49, main routing table version 49  
48 network entries using 6336 bytes of memory  
116 path entries using 6032 bytes of memory  
22/9 BGP path/bestpath attribute entries using 3696 bytes of memory  
18 BGP AS-PATH entries using 464 bytes of memory  
0 BGP route-map cache entries using 0 bytes of memory  
0 BGP filter-list cache entries using 0 bytes of memory  
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory  
BGP using 16560 total bytes of memory  
BGP activity 50/2 prefixes, 118/2 paths, scan interval 60 secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
48.73.240.13	4	1273	180	110	49	0	0	01:40:16	34
64.112.0.9	4	701	110	110	49	0	0	01:40:27	34
211.176.129.2	4	5511	181	110	49	0	0	01:40:38	34
211.176.129.6	4	1	93	110	49	0	0	01:39:23	0

AS 20717 - DE-CIX

```
! Router 16 =====
!  
R16#show ip bgp summary  
BGP router identifier 10.16.16.16, local AS number 20717
```

```

BGP table version is 368, main routing table version 368
48 network entries using 6336 bytes of memory
96 path entries using 4992 bytes of memory
17/9 BGP path/bestpath attribute entries using 2856 bytes of memory
14 BGP AS-PATH entries using 352 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 2) using 32 bytes of memory
BGP using 14568 total bytes of memory
BGP activity 58/10 prefixes, 325/229 paths, scan interval 60 secs

```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
46.88.20.1	4	5511	327	208	368	0	0	01:51:13	48
48.73.240.21	4	1273	326	192	368	0	0	02:28:40	48

AS 1 - Level 3 Parent, LLC

```

! BadGuy =====
!
BadGuy#show ip bgp summary
BGP router identifier 10.66.66.66, local AS number 1
BGP table version is 318, main routing table version 318
48 network entries using 6336 bytes of memory
48 path entries using 2496 bytes of memory
8/7 BGP path/bestpath attribute entries using 1344 bytes of memory
7 BGP AS-PATH entries using 184 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
BGP using 10360 total bytes of memory
BGP activity 67/19 prefixes, 138/90 paths, scan interval 60 secs

```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
211.176.129.5	4	4637	217	152	318	0	0	01:32:06	48

Prefixes received and sent from or to a determined neighbour

Example from Router 3 received and sent routes from / to Router 6

```

R3#show ip bgp neighbors 10.6.6.6 received-routes
BGP table version is 356, local router ID is 10.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

```

Network	Next Hop	Metric	LocPrf	Weight	Path
*>i46.87.162.0/24	10.6.6.6	0	100	0	5511 i
*>i46.87.162.110/32	10.6.6.6	0	100	0	5511 i
*>i46.87.162.111/32	10.6.6.6	0	100	0	5511 i
*>i46.87.162.112/32	10.6.6.6	0	100	0	5511 i
*>i46.88.20.0/24	10.6.6.6	0	100	0	5511 i
*>i46.88.22.0/24	10.6.6.6	0	100	0	5511 i
*>i46.89.36.0/24	10.6.6.6	0	100	0	5511 i
*>i46.89.38.0/24	10.6.6.6	0	100	0	5511 i
*>i46.92.2.0/24	10.6.6.6	0	100	0	5511 i
* i48.73.239.0/24	10.6.6.6	0	100	0	i
*>i48.73.239.66/32	10.6.6.6	0	100	0	i
* i48.73.240.0/22	10.6.6.6	0	100	0	i
* i63.25.64.0/18	10.6.6.6	0	100	0	i
*>i158.23.228.0/30	10.6.6.6	0	100	0	5511 23344 i
*>i158.23.228.0/24	10.6.6.6	0	100	0	5511 23344 i
*>i158.23.228.113/32	10.6.6.6	0	100	0	5511 23344 i
*>i211.176.128.0/30	10.6.6.6	0	100	0	4637 i
*>i211.176.128.0	10.6.6.6	0	100	0	4637 i
*>i211.176.128.115/32	10.6.6.6	0	100	0	4637 i
*>i211.176.129.0	10.6.6.6	0	100	0	4637 i
*>i211.176.130.0/23	10.6.6.6	0	100	0	4637 i
*>i211.176.132.0	10.6.6.6	0	100	0	4637 i
*>i211.176.135.0	10.6.6.6	0	100	0	4637 i
*>i211.176.136.0/22	10.6.6.6	0	100	0	4637 i

```
*>i211.176.137.0    10.6.6.6          0    100    0 4637 i
*>i211.176.138.0    10.6.6.6          0    100    0 4637 i
*>i211.176.139.0    10.6.6.6          0    100    0 4637 i
*>i211.176.140.0    10.6.6.6          0    100    0 4637 i
*>i211.176.141.0    10.6.6.6          0    100    0 4637 i
*>i211.176.142.0    10.6.6.6          0    100    0 4637 i
```

Total number of prefixes 30

R3#show ip bgp neighbors 10.6.6.6 advertised-routes

BGP table version is 356, local router ID is 10.3.3.3

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale

Origin codes: i - IGP, e - EGP, ? - incomplete

Network	Next Hop	Metric	LocPrf	Weight	Path
*>i46.87.162.0/24	10.6.6.6	0	100	0	5511 i
*>i46.87.162.110/32	10.6.6.6	0	100	0	5511 i
*>i46.87.162.111/32	10.6.6.6	0	100	0	5511 i
*>i46.87.162.112/32	10.6.6.6	0	100	0	5511 i
*>i46.88.20.0/24	10.6.6.6	0	100	0	5511 i
*>i46.88.22.0/24	10.6.6.6	0	100	0	5511 i
*>i46.89.36.0/24	10.6.6.6	0	100	0	5511 i
*>i46.89.38.0/24	10.6.6.6	0	100	0	5511 i
*>i46.92.2.0/24	10.6.6.6	0	100	0	5511 i
*>i48.73.239.0/30	10.2.2.2	0	100	0	i
*> 48.73.239.0/24	0.0.0.0	0		32768	i
*>i48.73.239.11/32	10.1.1.1	0	100	0	i
*>i48.73.239.22/32	10.2.2.2	0	100	0	i
*> 48.73.239.33/32	0.0.0.0	0		32768	i
*>i48.73.239.44/32	10.4.4.4	0	100	0	i
*>i48.73.239.55/32	10.5.5.5	0	100	0	i
*>i48.73.239.66/32	10.6.6.6	0	100	0	i
*> 48.73.240.0/22	0.0.0.0	0		32768	i
*> 48.73.240.4/30	0.0.0.0	0		32768	i
*> 63.25.64.0/18	0.0.0.0	0		32768	i
*>i64.96.0.0/30	10.5.5.5	0	100	0	701 i
*>i64.96.0.0/12	10.5.5.5	0	100	0	701 i
*>i64.96.0.114/32	10.5.5.5	0	100	0	701 i
*>i64.112.0.0/12	10.5.5.5	0	100	0	701 i
*>i65.0.204.0/24	10.5.5.5	0	100	0	701 i
*> 130.41.46.0/30	48.73.240.6			0	17390 i
*> 130.41.46.0/24	48.73.240.6	0		0	17390 i
*> 130.41.46.77/32	48.73.240.6	0		0	17390 i
*> 130.41.46.88/32	48.73.240.6			0	17390 i
*> 130.41.47.0/24	48.73.240.6			0	17390 64513 i
*> 130.41.47.99/32	48.73.240.6			0	17390 64513 i
*>i158.23.228.0/30	10.6.6.6	0	100	0	5511 23344 i
*>i158.23.228.0/24	10.6.6.6	0	100	0	5511 23344 i
*>i158.23.228.113/32	10.6.6.6	0	100	0	5511 23344 i
*>i211.176.128.0/30	10.6.6.6	0	100	0	4637 i
*>i211.176.128.0	10.6.6.6	0	100	0	4637 i
*>i211.176.128.115/32	10.6.6.6	0	100	0	4637 i
*>i211.176.129.0	10.6.6.6	0	100	0	4637 i
*>i211.176.130.0/23	10.6.6.6	0	100	0	4637 i
*>i211.176.132.0	10.6.6.6	0	100	0	4637 i
*>i211.176.135.0	10.6.6.6	0	100	0	4637 i
*>i211.176.136.0/22	10.6.6.6	0	100	0	4637 i
*>i211.176.137.0	10.6.6.6	0	100	0	4637 i
*>i211.176.138.0	10.6.6.6	0	100	0	4637 i
*>i211.176.139.0	10.6.6.6	0	100	0	4637 i
*>i211.176.140.0	10.6.6.6	0	100	0	4637 i
*>i211.176.141.0	10.6.6.6	0	100	0	4637 i
*>i211.176.142.0	10.6.6.6	0	100	0	4637 i

Total number of prefixes 48

Prefixes received and sent from or to a determined neighbour, notably the prefixes installed on the Loc-RIB BGP table

Example from Router 9

R9#show ip bgp

BGP table version is 358, local router ID is 10.9.9.9

Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
r RIB-failure, S Stale

Origin codes: i - IGP, e - EGP, ? - incomplete

Network	Next Hop	Metric	LocPrf	Weight	Path
* 46.87.162.0/24	130.41.46.9			0	17390 1273 5511 i
*>	130.41.46.5			0	17390 1273 5511 i
* 46.87.162.110/32	130.41.46.9			0	17390 1273 5511 i
*>	130.41.46.5			0	17390 1273 5511 i
*> 46.87.162.111/32	130.41.46.5			0	17390 1273 5511 i
*	130.41.46.9			0	17390 1273 5511 i
* 46.87.162.112/32	130.41.46.5			0	17390 1273 5511 i
*>	130.41.46.9			0	17390 1273 5511 i
* 46.88.20.0/24	130.41.46.9			0	17390 1273 5511 i
*>	130.41.46.5			0	17390 1273 5511 i
* 46.88.22.0/24	130.41.46.9			0	17390 1273 5511 i
*>	130.41.46.5			0	17390 1273 5511 i
* 46.89.36.0/24	130.41.46.9			0	17390 1273 5511 i
*>	130.41.46.5			0	17390 1273 5511 i
* 46.89.38.0/24	130.41.46.9			0	17390 1273 5511 i
*>	130.41.46.5			0	17390 1273 5511 i
* 46.92.2.0/24	130.41.46.9			0	17390 1273 5511 i
*>	130.41.46.5			0	17390 1273 5511 i
* 48.73.239.0/30	130.41.46.9			0	17390 1273 i
*>	130.41.46.5			0	17390 1273 i
* 48.73.239.0/24	130.41.46.9			0	17390 1273 i
*>	130.41.46.5			0	17390 1273 i
* 48.73.239.11/32	130.41.46.5			0	17390 1273 i
*>	130.41.46.9			0	17390 1273 i
* 48.73.239.22/32	130.41.46.9			0	17390 1273 i
*>	130.41.46.5			0	17390 1273 i
* 48.73.239.33/32	130.41.46.5			0	17390 1273 i
*>	130.41.46.9			0	17390 1273 i
* 48.73.239.44/32	130.41.46.9			0	17390 1273 i
*>	130.41.46.5			0	17390 1273 i
* 48.73.239.55/32	130.41.46.5			0	17390 1273 i
*>	130.41.46.9			0	17390 1273 i
* 48.73.239.66/32	130.41.46.9			0	17390 1273 i
*>	130.41.46.5			0	17390 1273 i
* 48.73.240.0/22	130.41.46.9			0	17390 1273 i
*>	130.41.46.5			0	17390 1273 i
* 48.73.240.4/30	130.41.46.5			0	17390 1273 i
*>	130.41.46.9			0	17390 1273 i
* 63.25.64.0/18	130.41.46.9			0	17390 1273 i
*>	130.41.46.5			0	17390 1273 i
* 64.96.0.0/30	130.41.46.5			0	17390 701 i
*>	130.41.46.9			0	17390 701 i
* 64.96.0.0/12	130.41.46.5			0	17390 701 i
*>	130.41.46.9			0	17390 701 i
* 64.96.0.114/32	130.41.46.5			0	17390 701 i
*>	130.41.46.9			0	17390 701 i
* 64.112.0.0/12	130.41.46.5			0	17390 701 i
*>	130.41.46.9			0	17390 701 i
* 65.0.204.0/24	130.41.46.5			0	17390 701 i
*>	130.41.46.9			0	17390 701 i
* 130.41.46.0/30	130.41.46.9			0	17390 i
*>	130.41.46.5	0		0	17390 i
* 130.41.46.0/24	130.41.46.9	0		0	17390 i
*>	130.41.46.5	0		0	17390 i
* 130.41.46.77/32	130.41.46.5			0	17390 i
*>	130.41.46.9	0		0	17390 i
* 130.41.46.88/32	130.41.46.9			0	17390 i
*>	130.41.46.5	0		0	17390 i
*> 130.41.47.0/24	0.0.0.0	0	32768		i
*> 130.41.47.99/32	0.0.0.0	0	32768		i
*> 158.23.228.0/30	130.41.46.5			0	17390 1273 5511 23344 i
*	130.41.46.9			0	17390 1273 5511 23344 i
*> 158.23.228.0/24	130.41.46.5			0	17390 1273 5511 23344 i
*	130.41.46.9			0	17390 1273 5511 23344 i
*> 158.23.228.113/32					

```

130.41.46.5      0 17390 1273 5511 23344 i
*                130.41.46.9      0 17390 1273 5511 23344 i
* 211.176.128.0/30 130.41.46.5      0 17390 1273 4637 i
*>                130.41.46.9      0 17390 701 4637 i
* 211.176.128.0    130.41.46.5      0 17390 1273 4637 i
*>                130.41.46.9      0 17390 701 4637 i
* 211.176.128.115/32
*                130.41.46.5      0 17390 1273 4637 i
*>                130.41.46.9      0 17390 701 4637 i
* 211.176.129.0    130.41.46.5      0 17390 1273 4637 i
*>                130.41.46.9      0 17390 701 4637 i
* 211.176.130.0/23 130.41.46.5      0 17390 1273 4637 i
*>                130.41.46.9      0 17390 701 4637 i
* 211.176.132.0    130.41.46.5      0 17390 1273 4637 i
*>                130.41.46.9      0 17390 701 4637 i
* 211.176.135.0    130.41.46.5      0 17390 1273 4637 i
*>                130.41.46.9      0 17390 701 4637 i
* 211.176.136.0/22 130.41.46.5      0 17390 1273 4637 i
*>                130.41.46.9      0 17390 701 4637 i
* 211.176.137.0    130.41.46.5      0 17390 1273 4637 i
*>                130.41.46.9      0 17390 701 4637 i
* 211.176.138.0    130.41.46.5      0 17390 1273 4637 i
*>                130.41.46.9      0 17390 701 4637 i
* 211.176.139.0    130.41.46.5      0 17390 1273 4637 i
*>                130.41.46.9      0 17390 701 4637 i
* 211.176.140.0    130.41.46.5      0 17390 1273 4637 i
*>                130.41.46.9      0 17390 701 4637 i
* 211.176.141.0    130.41.46.5      0 17390 1273 4637 i
*>                130.41.46.9      0 17390 701 4637 i
* 211.176.142.0    130.41.46.5      0 17390 1273 4637 i
*>                130.41.46.9      0 17390 701 4637 i

```

TCLSH script on the appendix TCLSH (Tool Control Language) Shell for testing

All routers passed the TCLSH ping tests with success, example from Router 12

```

R12#tclsh
R12(tcl)#foreach address {
+>48.73.239.11
+>48.73.239.22
+>48.73.239.33
+>48.73.239.44
+>48.73.239.55
+>48.73.239.66
+>130.41.46.77
+>130.41.46.88
+>130.41.47.99
+>46.87.162.110
+>46.87.162.111
+>46.87.162.112
+>158.23.228.113
+>64.96.0.114
+>211.176.128.115
+>130.41.46.1
+>48.73.239.1
+>64.96.0.1
+>211.176.128.1
+>46.87.162.1
+>158.23.228.1
+>} {ping $address source lo1}

```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 48.73.239.11, timeout is 2 seconds:

Packet sent with a source address of 46.87.162.112

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 60/81/132 ms

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 48.73.239.22, timeout is 2 seconds:

Packet sent with a source address of 46.87.162.112

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 40/49/60 ms

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 48.73.239.33, timeout is 2 seconds:

Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 40/53/64 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 48.73.239.44, timeout is 2 seconds:
Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/32/48 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 48.73.239.55, timeout is 2 seconds:
Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 48/67/92 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 48.73.239.66, timeout is 2 seconds:
Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/24/36 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 130.41.46.77, timeout is 2 seconds:
Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/63/76 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 130.41.46.88, timeout is 2 seconds:
Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 60/73/88 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 130.41.47.99, timeout is 2 seconds:
Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/80/104 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 46.87.162.110, timeout is 2 seconds:
Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/13/20 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 46.87.162.111, timeout is 2 seconds:
Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/12/16 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 46.87.162.112, timeout is 2 seconds:
Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/4 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 158.23.228.113, timeout is 2 seconds:
Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 16/42/84 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 64.96.0.114, timeout is 2 seconds:
Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 52/94/152 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 211.176.128.115, timeout is 2 seconds:
Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/26/52 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 130.41.46.1, timeout is 2 seconds:
Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 72/93/116 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 48.73.239.1, timeout is 2 seconds:
Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 56/72/96 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 64.96.0.1, timeout is 2 seconds:

```

Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 80/93/100 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 211.176.128.1, timeout is 2 seconds:
Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 32/44/64 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 46.87.162.1, timeout is 2 seconds:
Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 8/18/44 ms
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 158.23.228.1, timeout is 2 seconds:
Packet sent with a source address of 46.87.162.112
!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 12/39/84 ms

```

2.4 - Practical Questions

2.4.1 - How is the BGP next hop reachability solved inside an AS?

Inside an Autonomous System (AS), BGP next-hop reachability is resolved using the Interior Gateway Protocol (IGP), such as OSPF.

When an eBGP-learned route is advertised via iBGP, the next-hop IP address remains unchanged by default (it is the IP of the external peer).

To ensure that all routers within the AS can reach this next-hop IP, we use the next-hop-self command on iBGP sessions. This command changes the next-hop attribute to the router's own IP (usually the loopback address), which is already reachable via the IGP.

2.4.2 - Why is it a good practice to use the loopback IP address in the iBGP sessions?

Using a loopback interface for iBGP sessions offers:

- **Stability:** Loopback interfaces are always up unless administratively shut down, unlike physical interfaces.
- **Resilience:** If a physical link fails, the router can still reach the loopback via alternate paths.
- **Scalability:** Simplifies configuration in route reflector and peer-group setups.
- **Consistency:** Ensures the BGP session is not tied to a specific physical interface.

2.4.3 - Create and present a detailed table with all the connectivity tests performed using the TCLSH procedure, as previously outlined

AS	Source Router(s)	Destination IP	Result
AS 1273 - Vodafone	R1, R2, R3, R4, R5, R6	48.73.239.11 (R1 Lo1)	Success
		48.73.239.22 (R2 Lo1)	Success
		48.73.239.33 (R3 Lo1)	Success
		48.73.239.44 (R4 Lo1)	Success
		48.73.239.55 (R5 Lo1)	Success
		48.73.239.66 (R6 Lo1)	Success
		130.41.46.77 (R7 Lo1)	Success
		130.41.46.88 (R8 Lo1)	Success
		130.41.47.99 (R9 Lo1)	Success
		46.87.162.110 (R10 Lo1)	Success
		46.87.162.111 (R11 Lo1)	Success
		46.87.162.112 (R12 Lo1)	Success

		158.23.228.113 (R13 Lo1)	Success
		64.96.0.114 (R14 Lo1)	Success
		211.176.128.115 (R15 Lo1)	Success
		130.41.46.1 (Server1)	Success
		48.73.239.1 (Server2)	Success
		64.96.0.1 (Server3)	Success
		211.176.128.1 (Server4)	Success
		46.87.162.1 (Server5)	Success
		158.23.228.1 (Server6)	Success
AS 17390 - IBM	R7, R8	48.73.239.11 (R1 Lo1)	Success
		48.73.239.22 (R2 Lo1)	Success
		48.73.239.33 (R3 Lo1)	Success
		48.73.239.44 (R4 Lo1)	Success
		48.73.239.55 (R5 Lo1)	Success
		48.73.239.66 (R6 Lo1)	Success
		130.41.46.77 (R7 Lo1)	Success
		130.41.46.88 (R8 Lo1)	Success
		130.41.47.99 (R9 Lo1)	Success
		46.87.162.110 (R10 Lo1)	Success
		46.87.162.111 (R11 Lo1)	Success
		46.87.162.112 (R12 Lo1)	Success
		158.23.228.113 (R13 Lo1)	Success
		64.96.0.114 (R14 Lo1)	Success
		211.176.128.115 (R15 Lo1)	Success
		130.41.46.1 (Server1)	Success
		48.73.239.1 (Server2)	Success
		64.96.0.1 (Server3)	Success
		211.176.128.1 (Server4)	Success
		46.87.162.1 (Server5)	Success
		158.23.228.1 (Server6)	Success
AS 64513 - Private (inside IBM)	R9	48.73.239.11 (R1 Lo1)	Success
		48.73.239.22 (R2 Lo1)	Success
		48.73.239.33 (R3 Lo1)	Success
		48.73.239.44 (R4 Lo1)	Success
		48.73.239.55 (R5 Lo1)	Success
		48.73.239.66 (R6 Lo1)	Success
		130.41.46.77 (R7 Lo1)	Success
		130.41.46.88 (R8 Lo1)	Success

		130.41.47.99 (R9 Lo1)	Success
		46.87.162.110 (R10 Lo1)	Success
		46.87.162.111 (R11 Lo1)	Success
		46.87.162.112 (R12 Lo1)	Success
		158.23.228.113 (R13 Lo1)	Success
		64.96.0.114 (R14 Lo1)	Success
		211.176.128.115 (R15 Lo1)	Success
		130.41.46.1 (Server1)	Success
		48.73.239.1 (Server2)	Success
		64.96.0.1 (Server3)	Success
		211.176.128.1 (Server4)	Success
		46.87.162.1 (Server5)	Success
		158.23.228.1 (Server6)	Success
AS 5511 - Orange	R10, R11, R12	48.73.239.11 (R1 Lo1)	Success
		48.73.239.22 (R2 Lo1)	Success
		48.73.239.33 (R3 Lo1)	Success
		48.73.239.44 (R4 Lo1)	Success
		48.73.239.55 (R5 Lo1)	Success
		48.73.239.66 (R6 Lo1)	Success
		130.41.46.77 (R7 Lo1)	Success
		130.41.46.88 (R8 Lo1)	Success
		130.41.47.99 (R9 Lo1)	Success
		46.87.162.110 (R10 Lo1)	Success
		46.87.162.111 (R11 Lo1)	Success
		46.87.162.112 (R12 Lo1)	Success
		158.23.228.113 (R13 Lo1)	Success
		64.96.0.114 (R14 Lo1)	Success
		211.176.128.115 (R15 Lo1)	Success
		130.41.46.1 (Server1)	Success
		48.73.239.1 (Server2)	Success
		64.96.0.1 (Server3)	Success
		211.176.128.1 (Server4)	Success
		46.87.162.1 (Server5)	Success
		158.23.228.1 (Server6)	Success
AS 23344 - Disney	R13	48.73.239.11 (R1 Lo1)	Success
		48.73.239.22 (R2 Lo1)	Success
		48.73.239.33 (R3 Lo1)	Success

		48.73.239.44 (R4 Lo1)	Success
		48.73.239.55 (R5 Lo1)	Success
		48.73.239.66 (R6 Lo1)	Success
		130.41.46.77 (R7 Lo1)	Success
		130.41.46.88 (R8 Lo1)	Success
		130.41.47.99 (R9 Lo1)	Success
		46.87.162.110 (R10 Lo1)	Success
		46.87.162.111 (R11 Lo1)	Success
		46.87.162.112 (R12 Lo1)	Success
		158.23.228.113 (R13 Lo1)	Success
		64.96.0.114 (R14 Lo1)	Success
		211.176.128.115 (R15 Lo1)	Success
		130.41.46.1 (Server1)	Success
		48.73.239.1 (Server2)	Success
		64.96.0.1 (Server3)	Success
		211.176.128.1 (Server4)	Success
		46.87.162.1 (Server5)	Success
		158.23.228.1 (Server6)	Success
AS 701 - Verizon	R14	48.73.239.11 (R1 Lo1)	Success
		48.73.239.22 (R2 Lo1)	Success
		48.73.239.33 (R3 Lo1)	Success
		48.73.239.44 (R4 Lo1)	Success
		48.73.239.55 (R5 Lo1)	Success
		48.73.239.66 (R6 Lo1)	Success
		130.41.46.77 (R7 Lo1)	Success
		130.41.46.88 (R8 Lo1)	Success
		130.41.47.99 (R9 Lo1)	Success
		46.87.162.110 (R10 Lo1)	Success
		46.87.162.111 (R11 Lo1)	Success
		46.87.162.112 (R12 Lo1)	Success
		158.23.228.113 (R13 Lo1)	Success
		64.96.0.114 (R14 Lo1)	Success
		211.176.128.115 (R15 Lo1)	Success
		130.41.46.1 (Server1)	Success
		48.73.239.1 (Server2)	Success
		64.96.0.1 (Server3)	Success
		211.176.128.1 (Server4)	Success
		46.87.162.1 (Server5)	Success

		158.23.228.1 (Server6)	Success
AS 4637 - Telstra	R15	48.73.239.11 (R1 Lo1)	Success
		48.73.239.22 (R2 Lo1)	Success
		48.73.239.33 (R3 Lo1)	Success
		48.73.239.44 (R4 Lo1)	Success
		48.73.239.55 (R5 Lo1)	Success
		48.73.239.66 (R6 Lo1)	Success
		130.41.46.77 (R7 Lo1)	Success
		130.41.46.88 (R8 Lo1)	Success
		130.41.47.99 (R9 Lo1)	Success
		46.87.162.110 (R10 Lo1)	Success
		46.87.162.111 (R11 Lo1)	Success
		46.87.162.112 (R12 Lo1)	Success
		158.23.228.113 (R13 Lo1)	Success
		64.96.0.114 (R14 Lo1)	Success
		211.176.128.115 (R15 Lo1)	Success
		130.41.46.1 (Server1)	Success
		48.73.239.1 (Server2)	Success
		64.96.0.1 (Server3)	Success
		211.176.128.1 (Server4)	Success
		46.87.162.1 (Server5)	Success
		158.23.228.1 (Server6)	Success
AS 20717 - DE-CIX	R16 (without source Lo1)	48.73.239.11 (R1 Lo1)	Success
		48.73.239.22 (R2 Lo1)	Success
		48.73.239.33 (R3 Lo1)	Success
		48.73.239.44 (R4 Lo1)	Success
		48.73.239.55 (R5 Lo1)	Success
		48.73.239.66 (R6 Lo1)	Success
		130.41.46.77 (R7 Lo1)	Success
		130.41.46.88 (R8 Lo1)	Success
		130.41.47.99 (R9 Lo1)	Success
		46.87.162.110 (R10 Lo1)	Success
		46.87.162.111 (R11 Lo1)	Success
		46.87.162.112 (R12 Lo1)	Success
		158.23.228.113 (R13 Lo1)	Success
		64.96.0.114 (R14 Lo1)	Success
		211.176.128.115 (R15 Lo1)	Success
		130.41.46.1 (Server1)	Success

		48.73.239.1 (Server2)	Success
		64.96.0.1 (Server3)	Success
		211.176.128.1 (Server4)	Success
		46.87.162.1 (Server5)	Success
		158.23.228.1 (Server6)	Success
AS 1 - Level 3 Parent, LCC	BadGuy (without source Lo1)	48.73.239.11 (R1 Lo1)	Success
		48.73.239.22 (R2 Lo1)	Success
		48.73.239.33 (R3 Lo1)	Success
		48.73.239.44 (R4 Lo1)	Success
		48.73.239.55 (R5 Lo1)	Success
		48.73.239.66 (R6 Lo1)	Success
		130.41.46.77 (R7 Lo1)	Success
		130.41.46.88 (R8 Lo1)	Success
		130.41.47.99 (R9 Lo1)	Success
		46.87.162.110 (R10 Lo1)	Success
		46.87.162.111 (R11 Lo1)	Success
		46.87.162.112 (R12 Lo1)	Success
		158.23.228.113 (R13 Lo1)	Success
		64.96.0.114 (R14 Lo1)	Success
		211.176.128.115 (R15 Lo1)	Success
		130.41.46.1 (Server1)	Success
		48.73.239.1 (Server2)	Success
		64.96.0.1 (Server3)	Success
		211.176.128.1 (Server4)	Success
		46.87.162.1 (Server5)	Success
		158.23.228.1 (Server6)	Success

2.4.4 - In the following Local-RIB table output example, how many routes were installed for the destination 46.87.162.0/24? Justify your response explaining the decision process in BGP

Network	Next Hop	Metric	LocPrf	Weight	Path
* 46.87.162.0/24	130.41.46.5			0 17390	1273 20717 5511 i
*>	130.41.46.9			0 17390	1273 20717 5511 i

Only one route is installed in the routing table — the one marked with *>.

The * indicates a valid route, and > indicates the best path.

BGP selects only one best path per prefix based on its decision process (Weight → Local Preference → AS Path → etc.).

Here, both paths have the same attributes, so the tie-breaker is likely the lowest router ID or lowest neighbor IP.

Only the best path is installed in the routing table and advertised to other BGP peers.

2.4.5 - What would have happened in case you didn't configure the next-hop-self on the iBGP peering definitions? What reasons explain why BGP doesn't set the next-hop-self as a default setting?

If `next-hop-self` is not configured:

- The next-hop IP of eBGP-learned routes remains the external peer's IP.

- Internal routers (iBGP peers) may not have a route to that external IP in their routing table.
- This leads to unreachable next-hop and the BGP route is not used.

Why it's not default:

BGP assumes that the next-hop IP is reachable via the IGP. In some designs (e.g., MPLS VPNs, or when using static routes), the next-hop may be intentionally preserved.

Making ``next-hop-self`` the default could break such designs, so it is left as an explicit configuration choice.

2.4.6 - How are the route prefixes propagated inside the AS when you have route reflectors configured?

In a route reflector (RR) setup:

- The RR reflects iBGP-learned routes to its clients and other RRs.
- Clients only peer with the RR, not with each other (reduces full-mesh requirement).
- The RR preserves BGP attributes like NEXT_HOP, AS_PATH, and LOCAL_PREF.
- Routes learned from an eBGP peer are reflected to all iBGP clients and other RRs.
- Routes learned from a client are reflected to all other clients and non-client peers (including eBGP).

This allows scalable propagation of prefixes across the AS without a full iBGP mesh.

2.4.7 - Simulate and explain using Wireshark the BGP messages associated to each BGP state (see: [article](#))

BGP uses a state machine with the following states and associated messages:

BGP State	Message Type	Description
Idle	–	Initial state; no resources allocated.
Connect	TCP SYN	Attempts to establish TCP connection (port 179).
Active	TCP SYN	Actively tries to connect if passive fails.
OpenSent	OPEN	TCP established; sends BGP OPEN message.
OpenConfirm	KEEPALIVE	Waits for KEEPALIVE; if received, moves to Established.
Established UPDATE Session is up; routes are exchanged via UPDATE messages.		

No.	Time	Source	Destination	Protocol	Length	Info
6	6.764150	48.73.240.14	48.73.240.13	TCP	60	56864 → 179 [SYN] Seq=0 Win=16384 Len=0 MSS=1460
7	6.778882	48.73.240.13	48.73.240.14	TCP	60	179 → 56864 [SYN, ACK] Seq=0 Ack=1 Win=16384 Len=0 MSS=1460
8	6.794579	48.73.240.14	48.73.240.13	TCP	60	56864 → 179 [ACK] Seq=1 Ack=1 Win=16384 Len=0
9	6.794579	48.73.240.14	48.73.240.13	BGP	107	OPEN Message
10	6.809702	48.73.240.13	48.73.240.14	BGP	107	OPEN Message
11	6.809702	48.73.240.13	48.73.240.14	BGP	73	KEEPALIVE Message
12	6.825730	48.73.240.14	48.73.240.13	BGP	73	KEEPALIVE Message
13	7.039386	48.73.240.13	48.73.240.14	TCP	60	179 → 56864 [ACK] Seq=73 Ack=73 Win=16312 Len=0
24	63.580744	48.73.240.13	48.73.240.14	BGP	73	KEEPALIVE Message
25	63.966439	48.73.240.14	48.73.240.13	TCP	60	56864 → 179 [ACK] Seq=73 Ack=92 Win=16293 Len=0
27	73.103132	48.73.240.13	48.73.240.14	BGP	116	UPDATE Message
28	73.488933	48.73.240.14	48.73.240.13	TCP	60	56864 → 179 [ACK] Seq=73 Ack=154 Win=16231 Len=0
34	103.504042	48.73.240.13	48.73.240.14	BGP	180	UPDATE Message, UPDATE Message
35	103.858362	48.73.240.14	48.73.240.13	TCP	60	56864 → 179 [ACK] Seq=73 Ack=280 Win=16105 Len=0
42	134.229093	48.73.240.13	48.73.240.14	BGP	109	UPDATE Message
43	134.668389	48.73.240.14	48.73.240.13	TCP	60	56864 → 179 [ACK] Seq=73 Ack=335 Win=16050 Len=0
44	135.745729	48.73.240.14	48.73.240.13	BGP	432	UPDATE Message, UPDATE Message, UPDATE Message, UPDATE Message, UPDATE Message, UPDATE Message
45	135.745729	48.73.240.14	48.73.240.13	BGP	73	KEEPALIVE Message
46	135.961059	48.73.240.13	48.73.240.14	TCP	60	179 → 56864 [ACK] Seq=335 Ack=470 Win=15915 Len=0
51	164.935044	48.73.240.13	48.73.240.14	BGP	304	UPDATE Message, UPDATE Message, UPDATE Message, UPDATE Message
52	165.381771	48.73.240.14	48.73.240.13	TCP	60	56864 → 179 [ACK] Seq=470 Ack=585 Win=15800 Len=0
60	202.963788	48.73.240.14	48.73.240.13	BGP	176	UPDATE Message, UPDATE Message
61	203.195993	48.73.240.13	48.73.240.14	TCP	60	179 → 56864 [ACK] Seq=585 Ack=592 Win=15793 Len=0
65	225.367711	48.73.240.13	48.73.240.14	BGP	73	KEEPALIVE Message
66	225.846683	48.73.240.14	48.73.240.13	TCP	60	56864 → 179 [ACK] Seq=592 Ack=604 Win=15781 Len=0
78	285.748848	48.73.240.13	48.73.240.14	BGP	73	KEEPALIVE Message
79	286.178261	48.73.240.14	48.73.240.13	TCP	60	56864 → 179 [ACK] Seq=592 Ack=623 Win=15762 Len=0
89	337.697942	48.73.240.14	48.73.240.13	BGP	73	KEEPALIVE Message
90	337.941241	48.73.240.13	48.73.240.14	TCP	60	179 → 56864 [ACK] Seq=623 Ack=611 Win=15774 Len=0
92	346.199423	48.73.240.13	48.73.240.14	BGP	73	KEEPALIVE Message
93	346.658448	48.73.240.14	48.73.240.13	TCP	60	56864 → 179 [ACK] Seq=611 Ack=642 Win=15743 Len=0

BGP Messages

- ✓ **Border Gateway Protocol - UPDATE Message**
 - Marker: ffffffffffffffffffffffffffffffffff
 - Length: 59
 - Type: UPDATE Message (2)
 - Withdrawn Routes Length: 0
 - Total Path Attribute Length: 24
 - > Path attributes
 - ✓ **Network Layer Reachability Information (NLRI)**
 - > 211.176.132.0/24
 - > 211.176.135.0/24
 - > 211.176.136.0/22
- ✓ **Border Gateway Protocol - UPDATE Message**
 - Marker: ffffffffffffffffffffffffffffffffff
 - Length: 69
 - Type: UPDATE Message (2)
 - Withdrawn Routes Length: 0
 - Total Path Attribute Length: 38
 - > Path attributes
 - ✓ **Network Layer Reachability Information (NLRI)**
 - > 211.176.140.0/23
 - > 211.176.128.0/22
- ✓ **Border Gateway Protocol - UPDATE Message**
 - Marker: ffffffffffffffffffffffffffffffffff
 - Length: 57
 - Type: UPDATE Message (2)
 - Withdrawn Routes Length: 0
 - Total Path Attribute Length: 24
 - > Path attributes
 - ✓ **Network Layer Reachability Information (NLRI)**
 - > 64.96.0.0/12
 - > 64.112.0.0/12
 - > 65.0.204.0/24

Advertised routes in **UPDATE**

- OPEN: Contains AS number, BGP version, hold time, and BGP identifier.
- KEEPALIVE: Maintains session alive.
- UPDATE: Advertises or withdraws routes.
- NOTIFICATION: Used for error reporting or session termination.

3 - Phase 3 - Routing Policies Implementation

3.1 - Objectives

In this phase, we have as objectives:

- Aggregate IP prefixes when possible and advertise only the aggregate route on eBGP sessions
- We should not have advertised prefixes longer than /24 in the internet routing table
- The internet peering's should accept a maximum of 50 prefixes
- We should not have private ASs in the middle from an AS path attribute
- The AS 23344 has only one peering to the internet and want to receive only the default route from the eBGP peering

3.2 - Implementation

In the ASes with one peer router only and that advertise routes, we used the `aggregate-address` command.

Example from Router 9

```
! AS 64513 (private to IBM) - Router 9 =====
router bgp 64513
  bgp router-id 10.9.9.9
  ! ...
  network 130.41.47.0 mask 255.255.255.0
  aggregate-address 130.41.47.0 255.255.255.0 summary-only
```

On the ASes with more than one router, we created an access list restricting the prefixes to advertise to the eBGP peers.

Example from Router 1

```
! AS 1273 Vodafone =====
! All routers -----
ip prefix-list PREFIX_LIST_ADV_SUMM seq 5 permit 0.0.0.0/0 le 24
!
! Router 1 -----
router bgp 1273
  bgp router-id 10.1.1.1
  ! ...
  neighbor 48.73.240.2 remote-as 17390
  neighbor 48.73.240.2 soft-reconfiguration inbound
  neighbor 48.73.240.2 prefix-list PREFIX_LIST_ADV_SUMM out
```

To limit the internet peering's to a maximum of 50 prefixes, in all the BGP's `neighbour`, we use the command `neighbor [peer IP address] maximum-prefix 50`

To avoid the private AS (AS 64513) to be announced outside to the internet, in the routers R7 and R8, we used the command: `neighbor [peer IP address] remove-private-as`. This prevented the advertising of the network announced by the router R9 (AS 64513) outside the AS 17390.

The AS 23344 has only one peering to the internet and want to receive only the default route from the eBGP peering. To accomplish that, in the router R11 that belongs to the only AS (AS 5511 - Orange) that connects to R13 of AS 23344 - Disney, we used the command `neighbor 46.88.20.6 default-originate`

3.3 - Test and Validation

Inspection of the `AS_PATH` attribute in each router

To accomplish this, we use the commands:

- `show ip bgp 211.176.129.0`
- `show ip bgp`

The private AS 64515 is not announced to the internet, only to the AS 17390 IBM (routers R7 and R8). As an output example, we annexed the output of the router R3:

```
R3#show ip bgp 211.176.129.0
BGP routing table entry for 211.176.128.0/22, version 32
Paths: (1 available, best #1, table Default-IP-Routing-Table)
Flag: 0x820
  Advertised to update-groups:
    1 2 4
  4637, (aggregated by 4637 10.15.15.15), (Received from a RR-client), (received & used)
    10.6.6.6 (metric 3) from 10.6.6.6 (10.6.6.6)
      Origin IGP, metric 0, localpref 100, valid, internal, atomic-aggregate, best
```

```
R3#show ip bgp
BGP table version is 32, local router ID is 10.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete
```

Network	Next Hop	Metric	LocPrf	Weight	Path
*>i46.87.162.0/24	10.6.6.6	0	100	0	5511 i
*>i46.88.20.0/24	10.6.6.6	0	100	0	5511 i
*>i46.88.22.0/24	10.6.6.6	0	100	0	5511 i
*>i46.89.36.0/24	10.6.6.6	0	100	0	5511 i
*>i46.89.38.0/24	10.6.6.6	0	100	0	5511 i
*>i46.92.2.0/24	10.6.6.6	0	100	0	5511 i
*>i48.73.239.0/30	10.2.2.2	0	100	0	i
* i48.73.239.0/24	10.2.2.2	0	100	0	i
* i	10.4.4.4	0	100	0	i
* i	10.5.5.5	0	100	0	i
* i	10.1.1.1	0	100	0	i
* i	10.6.6.6	0	100	0	i
*>	0.0.0.0	0		32768	i
*>i48.73.239.11/32	10.1.1.1	0	100	0	i
*>i48.73.239.22/32	10.2.2.2	0	100	0	i
*> 48.73.239.33/32	0.0.0.0	0		32768	i
*>i48.73.239.44/32	10.4.4.4	0	100	0	i
*>i48.73.239.55/32	10.5.5.5	0	100	0	i
*>i48.73.239.66/32	10.6.6.6	0	100	0	i
* i48.73.240.0/22	10.2.2.2	0	100	0	i
* i	10.4.4.4	0	100	0	i
* i	10.5.5.5	0	100	0	i
* i	10.1.1.1	0	100	0	i
* i	10.6.6.6	0	100	0	i
*>	0.0.0.0	0		32768	i
*> 48.73.240.4/30	0.0.0.0	0		32768	i
* i63.25.64.0/18	10.2.2.2	0	100	0	i
* i	10.4.4.4	0	100	0	i
* i	10.5.5.5	0	100	0	i
* i	10.1.1.1	0	100	0	i
* i	10.6.6.6	0	100	0	i
*>	0.0.0.0	0		32768	i
* 64.96.0.0/24	48.73.240.6			0 17390	701 i
*>i	10.5.5.5	0	100	0	701 i
* 64.96.0.0/12	48.73.240.6			0 17390	701 i
*>i	10.5.5.5	0	100	0	701 i
* 64.112.0.0/12	48.73.240.6			0 17390	701 i
*>i	10.5.5.5	0	100	0	701 i
* 65.0.204.0/24	48.73.240.6			0 17390	701 i
*>i	10.5.5.5	0	100	0	701 i
*> 130.41.46.0/24	48.73.240.6	0		0 17390	i
* i	10.1.1.1	0	100	0	17390 i
*> 130.41.47.0/24	48.73.240.6			0 17390	i
*>i158.23.228.0/24	10.6.6.6	0	100	0	5511 23344 i
*>i211.176.128.0/22	10.6.6.6	0	100	0	4637 i
*>i211.176.132.0	10.6.6.6	0	100	0	4637 i
*>i211.176.135.0	10.6.6.6	0	100	0	4637 i
*>i211.176.136.0/22	10.6.6.6	0	100	0	4637 i
*>i211.176.140.0/23	10.6.6.6	0	100	0	4637 i

The output of router 8, with the private AS advertised:

```
R8# Show ip bgp
...

* i130.41.47.0/24 10.7.7.7 0 100 0 64513 i
```



```
*> 130.41.46.6 0 0 64513 i
...
```

3.4 - Practical Questions

3.4.1 - After applying your prefix-list or route-map, compare the output of `show ip bgp` and `show ip bgp neighbors <ip> advertised-routes`. Explain the differences you observe referring to the Adj-RIB-In, Adj-RIB-Out, and Loc-RIB tables

In our lab, we applied the following outbound prefix filter to eBGP peers to ensure only prefixes with a mask of /24 or less are advertised:

```
ip prefix-list PREFIX_LIST_ADV_SUMM seq 5 permit 0.0.0.0/0 le 24
```

This prefix list permits any prefix with a subnet mask length ≤ 24 and implicitly denies longer prefixes (e.g., /32 loopback addresses).

Comparison of Command Outputs:

- `show ip bgp` (Loc-RIB Table)
This command displays all BGP routes that are valid and installed in the local BGP table after inbound policies have been applied. It includes:
 - All prefixes learned from all peers (both internal and external).
 - Prefixes with masks longer than /24 (e.g., /32 Loopback1 addresses).
 - All candidate routes for best-path selection.
- `show ip bgp neighbors <ip> advertised-routes` (Adj-RIB-Out Table)
This command shows only the routes that have passed outbound policy filters and will be advertised to the specified neighbor. After applying PREFIX_LIST_ADV_SUMM, this table will:
 - Exclude all /32 prefixes (Loopback1 addresses).
 - Include only prefixes with masks /24 or shorter (e.g., server subnets like 46.87.162.0/24).
 - Reflect the effect of any summary-only aggregate commands.

Example:

In `show ip bgp`, we see:

```
48.73.239.11/32
46.87.162.0/24
```

In `show ip bgp neighbors 48.73.240.6 advertised-routes`, we see:

```
46.87.162.0/24
```

But not 48.73.239.11/32.

BGP Table Relationship Summary:

- **Adj-RIB-In:** Routes received from neighbor → **before** inbound policy.
- **Loc-RIB:** After inbound policy + best-path selection → **used for local routing**.
- **Adj-RIB-Out:** After outbound policy → what is **advertised to neighbor**.

So, our prefix list PREFIX_LIST_ADV_SUMM acts as an **outbound filter** that removes longer prefixes (like /32 loopbacks) from the **Adj-RIB-Out** table, ensuring only summarized or appropriately sized prefixes are advertised to external peers. This aligns with BGP best practices to reduce routing table size and improve scalability.

3.4.2 - Why is the control from the number of prefixes advertised to the internet a good practice?

Controlling the number of prefixes advertised is important for:

- **Scalability:** Prevents the global BGP table from growing excessively, which would increase memory and CPU usage on routers.
- **Stability:** Reduces the risk of route flapping and BGP convergence issues.

- **Security:** Limits the impact of route leaks or hijacks.
- **Policy compliance:** Many ISPs enforce prefix limits in peering agreements to ensure efficient routing.

In this lab, limiting prefixes to a maximum of 50 per peer helps maintain a manageable and stable routing environment

3.4.3 - What is the private AS number range in BGP? Describe some scenarios where using private ASNs can be useful.

Private ASN Ranges:

Format	Private Range
2-byte	64512 to 65534
4-byte	4200000000 to 4294967294

Use cases for private ASNs:

- **Internal network segmentation:** Within a larger AS, private ASNs can be used for testing or staging environments without consuming public AS numbers.
- **Mergers and migrations:** During network integration, private ASNs can be used temporarily to avoid conflicts.
- **Customer networks:** In some BGP designs, customers may use a private ASN for their internal routing, which is removed before advertisement to the global Internet (as done in this lab with AS 17390).

3.4.4 - Assume that you successfully configured prefix filtering. Based on the BGP path selection rules, explain why R12 selects a specific path for the prefix 65.0.204.0/24. Use the following output as reference [link](#):

```
R12# sh ip bgp 65.0.204.0
BGP routing table entry for 65.0.204.0/24, version 18717
Paths: (2 available, best #2, table Default-IP-Routing-Table)
  Advertised to update-groups:
    2
  4637 701, (received & used)
    10.10.10.10 (metric 2) from 10.10.10.10 (10.10.10.10)
      Origin IGP, metric 0, localpref 100, valid, internal
  1273 701, (received & used)
    48.73.240.17 from 48.73.240.17 (10.6.6.6)
      Origin IGP, localpref 100, valid, external, best
```

R12 selects the path via **1273 701** as best because:

1. Both paths have the same **Local Preference (100)**.
2. The path via **1273 701** is **external** (eBGP), while the other is **internal** (iBGP). BGP prefers **eBGP over iBGP** when other attributes are equal.
3. The **AS_PATH** length is the same (2 AS hops).
4. No difference in **MED, Origin**, or other attributes to influence the decision.

3.4.5 - Imagine that one of your prefixes was not selected as the best path in your lab. Based on the BGP decision process, propose a configuration change (e.g., adjusting local preference, MED, or weight) that would alter the selection. Justify your answer by showing the relevant command(s) and predicting the impact on the routing table.

Supposing that prefix `46.88.20.0/24` is not selected as best because it comes via an AS with longer **AS_PATH**. To influence selection, we could increase the **Local Preference** for that path.

Configuration change on R12 (inbound from desired neighbor):

```
route-map PREFER_PATH permit 10
  match ip address prefix-list PREFIX_LIST
  set local-preference 200
!
router bgp 5511
  neighbor 48.73.240.18 route-map PREFER_PATH in
```

Increasing **Local Preference** to 200 makes this path more preferred than others with default **Local Preference** (100). This change influences the first step of BGP decision process, ensuring this path is selected as best.

4 - Phase 4 - Influence the Internet Routing

4.1 - Objectives

In this phase the objectives are:

- Use BGP attributes to ensure **AS20717** is the preferred path for traffic originated in **AS1273** and **AS511**
- Enforce this preference symmetrically, guaranteeing consistent selection in both directions.
- Apply the rule along the full path (e.g., AS17390 <-> AS1273 <-> IXP1 <-> AS551) for routing coherence

4.2 - Implementation

To ensure that **AA20717** is the preferred path for traffic originated in **AS1273** with destination or passing through **AS511**, or traffic originated in **AS511** with destination or passing through **AS1273**, we first created an AS-Path Access List that matches the desired route.

```
Router6 - AS1273 =====
ip as-path access-list 1 permit _20717_5511$

Router10 - AS5511 =====
ip as-path access-list 1 permit _20171_1273$
```

Then, we created route map with `weight 300` matching the previously configured AS-Path ACL, in both routers (R6 - AS1273 - Vodafone, and R10 - AS5511 - Orange)

```
Router6 - AS1273 =====
route-map RM_PREF_LOCAL_AS5511 permit 10
  match as-path 1          ! Match routes with AS-PATH _20717_5511$
  set weight 300           ! Set high weight to prefer these routes

Router10 - AS5511 =====
route-map RM_PREF_LOCAL_AS1273 permit 10
  match as-path 1          ! Match routes with AS-PATH _20171_1273$
  set weight 300           ! Set high weight to prefer these routes
```

At least, in the direct links connecting the transit **AS20717** - DE-CIX, in the both routers we apply the Route Map

```
Router6 - AS1273 =====
router bgp 1273
! ...
neighbor 48.73.240.22 remote-as 20717
! ...
neighbor 48.73.240.22 route-map RM_PREF_LOCAL_AS5511 in
! ...

Router10 - AS5511 =====
router bgp 5511
! ...
neighbor 46.88.20.2 remote-as 20717
! ...
neighbor 46.88.20.2 route-map RM_PREF_LOCAL_AS1273 in
```

4.3 - Test and Validation

Routes through AS20717

Router 6

```
R6#sh ip bgp regexp _20717_
BGP table version is 41, local router ID is 10.6.6.6
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete
```

```

      Network      Next Hop      Metric LocPrf Weight Path
*> 46.87.162.0/24  48.73.240.22                300 20717 5511 i
*> 46.88.20.0/24   48.73.240.22                300 20717 5511 i
*> 46.88.22.0/24   48.73.240.22                300 20717 5511 i
*> 46.89.36.0/24   48.73.240.22                300 20717 5511 i
*> 46.89.38.0/24   48.73.240.22                300 20717 5511 i
*> 46.92.2.0/24    48.73.240.22                300 20717 5511 i

R6#show ip bgp neighbors 48.73.240.22 advertised-routes
BGP table version is 41, local router ID is 10.6.6.6
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

      Network      Next Hop      Metric LocPrf Weight Path
*> 46.87.162.0/24  48.73.240.22                300 20717 5511 i
*> 46.88.20.0/24   48.73.240.22                300 20717 5511 i
*> 46.88.22.0/24   48.73.240.22                300 20717 5511 i
*> 46.89.36.0/24   48.73.240.22                300 20717 5511 i
*> 46.89.38.0/24   48.73.240.22                300 20717 5511 i
*> 46.92.2.0/24    48.73.240.22                300 20717 5511 i
...

Total number of prefixes 22

R6#trace 46.87.162.110

Type escape sequence to abort.
Tracing the route to 46.87.162.110

  1 48.73.240.22 28 msec 36 msec 12 msec
  2 46.88.20.1 [AS 5511] 12 msec 20 msec *
```

Router 10

```

R10#sh ip bgp regexp _20717_
BGP table version is 36, local router ID is 10.10.10.10
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

      Network      Next Hop      Metric LocPrf Weight Path
*> 48.73.239.0/24  46.88.20.2                300 20717 1273 i
*> 48.73.240.0/22  46.88.20.2                300 20717 1273 i
*> 63.25.64.0/18   46.88.20.2                300 20717 1273 i

R10#show ip bgp neighbors 46.88.20.2 advertised-routes
BGP table version is 36, local router ID is 10.10.10.10
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

      Network      Next Hop      Metric LocPrf Weight Path
...
*> 48.73.239.0/24  46.88.20.2                300 20717 1273 i
*> 48.73.240.0/22  46.88.20.2                300 20717 1273 i
*> 63.25.64.0/18   46.88.20.2                300 20717 1273 i
...

Total number of prefixes 22

R10#trace 48.73.239.66

Type escape sequence to abort.
Tracing the route to 48.73.239.66

  1 46.88.20.2 40 msec 12 msec 12 msec
  2 48.73.240.21 [AS 1273] 12 msec 36 msec *
```

4.4 - Practical Questions

4.4.1 - Describe in your report the policy options you used to implement the routing policies (include all the details of the configurations (e.g., prefix-list, route-map, etc))

Described and explained in the previous point 4.2 - Implementation

4.4.2 - Provide command output screenshots that demonstrate the successful application of the configured policies

Described and explained in the previous point 4.3 - Test and Validation

4.4.3 - Discuss other alternative to achieve the same results and comment on their relative pros and cons, compared to your implementation

Alternative Approaches to Achieve the Same Result:

1. Local Preference (LOCAL_PREF)

```
! Instead of WEIGHT (Cisco proprietary, local to router)
route-map RM_PREF_LOCAL_AS5511 permit 10
  match as-path 1
  set local-preference 200          ! Higher LOCAL_PREF wins (default is 100)
```

Pros:

- Standard BGP attribute (RFC 4271) - interoperable across vendors
- Propagated within the AS to all iBGP peers
- More granular control (can be set on per-neighbor basis)
- Well-understood and widely deployed

Cons:

- Affects all routers in the AS (global effect within the AS)
- Requires careful planning to avoid unintended consequences
- Cannot override LOCAL_PREF from other iBGP speakers within same AS

Comparison to our WEIGHT implementation:

- WEIGHT is router-local, LOCAL_PREF is AS-wide
- WEIGHT has higher precedence than LOCAL_PREF in Cisco's decision process
- Our WEIGHT approach is simpler for single-router policy but doesn't scale to multi-router AS

2. AS_PATH Prepending

```
! Make other paths less desirable by artificially lengthening AS_PATH
route-map RM_PREPEND_OUT permit 10
  set as-path prepend 1273 1273 1273 ! Add our AS 3 times
```

Applied to outbound advertisements to other transit providers (not DE-CIX).

Pros:

- Simple to implement and understand
- Effective for making your AS less preferred by others
- Works well for outbound traffic engineering
- No special configuration needed on neighbor's side

Cons:

- Only affects inbound traffic to your AS
- Can be filtered by neighbors with AS_PATH length limits
- Less precise control compared to WEIGHT/LOCAL_PREF
- Doesn't help with our specific requirement (preferring certain transit paths)

3. MED (Multi-Exit Discriminator)

```
! Lower MED is preferred (default is 0)
route-map RM_SET_MED permit 10
  match as-path 1
  set metric 50
! Lower MED for preferred paths
```

Pros:

- Standard BGP attribute
- Useful when you have multiple connections to same AS
- Can influence neighboring AS's routing decisions

Cons:

- Only comparable between routes from the same neighboring AS
- Often ignored by many ISPs for security/policy reasons
- Not transitive (only sent to direct neighbors)
- Less effective for our cross-AS path preference scenario

4. Community-Based Traffic Engineering

```
! Tag routes with communities and apply policies based on them
route-map RM_TAG_COMMUNITIES permit 10
  match as-path 1
  set community 1273:100 20717:200 ! Tag with custom communities

! Neighbor configuration to send communities
neighbor 48.73.240.22 send-community
neighbor 48.73.240.22 route-map RM_TAG_COMMUNITIES out
```

Pros:

- Highly scalable and flexible
- Enables coordinated policies across multiple ASes
- Can implement sophisticated routing policies
- Well-supported by major ISPs

Cons:

- Requires agreement and coordination between ASes
- More complex to implement and debug
- Dependent on neighbor honoring your communities
- Not all ISPs accept or act on customer communities

5. BGP Add-Path (RFC 7911)

```
! Enable multiple paths for same prefix
router bgp 1273
  bgp additional-paths select best 3
  neighbor 48.73.240.22 additional-paths send receive
```

Pros:

- Maintains multiple paths simultaneously
- Enables faster failover
- More granular path selection capabilities
- Future-proof approach

Cons:

- Requires support on both ends
- Increased memory and CPU usage
- More complex configuration
- Not yet universally deployed

5 - Phase 5 - Security Practices

5.1 Objectives

In this last phase, the objectives are:

- Activate the MD5 authentication on all the eBGP peering's on the AS701. This a good practice for all the peers nevertheless for the lab proposes will be ok to test on this AS only
- Filter the `Bogon` prefixes on the peer routers from AS 1273. The BadGuy router is advertising `Bogons` (see: [BGP Best Practices for ISPs](#))
- Implement Remote Triggered Black Hole (RTBH) filtering - a popular and effective technique for the mitigation of denial-of-service (DoS) attack on AS1273 coming from the prefix `63.96.0.115` . [Remotely Triggered Black Hole Filtering - Destination Based and Source Based](#)

5.2 - Implementation

Enable MD5 Authentication on eBGP Peerings (AS701)

In all the routers (R5, R7, R14 and R15) we used the command `service password-encryption` , to prevent the visualisation of the password when running the command `show running-configuration`

On router R14 from AS 701 - Verizon

```
router bgp 701
  bgp router-id 10.14.14.14
  ! ...
  neighbor 64.112.0.10 remote-as 4637
  neighbor 64.112.0.10 password BGP@p455w0rd_701-4637
  ! ...
  neighbor 64.112.0.2 remote-as 1273
  neighbor 64.112.0.2 password BGP@p455w0rd_701-1273
  ! ...
  neighbor 64.112.0.6 remote-as 17390
  neighbor 64.112.0.6 password BGP@p455w0rd_701-17390
  ! ...
```

On Router R5 from AS 1273 - Vodafone

```
router bgp 1273
  bgp router-id 10.5.5.5
  ! ...
  neighbor 64.112.0.1 remote-as 701
  neighbor 64.112.0.1 password BGP@p455w0rd_701-1273
  ! ...
```

On router R7 from AS 17390 IBM

```
router bgp 17390
  bgp router-id 10.7.7.7
  ! ...
  neighbor 64.112.0.5 remote-as 701
  neighbor 64.112.0.5 password BGP@p455w0rd_701-17390
  ! ...
```

On router R15 from AS 4637 Telstra

```
router bgp 4637
  bgp router-id 10.15.15.15
  !...
  neighbor 64.112.0.9 remote-as 701
  neighbor 64.112.0.9 password BGP@p455w0rd_701-4637
  ! ...
```

Filter Bogon Prefixes from BadGuy Router

In all the routers, we defined a **BOGON-FILTER** with all the IP addresses that are not permitted in the internet

```
ip prefix-list BOGON-FILTER seq 5 deny 0.0.0.0/8 le 32      ! Software (current local network)
ip prefix-list BOGON-FILTER seq 10 deny 10.0.0.0/8 le 32    ! Private
ip prefix-list BOGON-FILTER seq 15 deny 100.64.0.0/10 le 32  ! Shared Address Space - CG-NAT
ip prefix-list BOGON-FILTER seq 20 deny 127.0.0.0/8 le 32   ! Loopback addresses to Localhost
ip prefix-list BOGON-FILTER seq 25 deny 169.254.0.0/16 le 32 ! APIPA - Automatic Private IP Address
ip prefix-list BOGON-FILTER seq 30 deny 172.16.0.0/12 le 32  ! Private
ip prefix-list BOGON-FILTER seq 35 deny 192.0.0.0/24 le 32   ! IETF Protocol Assignments
ip prefix-list BOGON-FILTER seq 40 deny 192.0.2.0/24 le 32   ! Documentation - TEST-NET-1
ip prefix-list BOGON-FILTER seq 45 deny 192.88.99.0/24 le 32 ! Reserved
ip prefix-list BOGON-FILTER seq 50 deny 192.168.0.0/16 le 32 ! Private
ip prefix-list BOGON-FILTER seq 55 deny 198.18.0.0/15 le 32  ! Benchmark testing
ip prefix-list BOGON-FILTER seq 60 deny 198.51.100.0/24 le 32 ! Documentation - TEST-NET-2
ip prefix-list BOGON-FILTER seq 65 deny 203.0.113.0/24 le 32 ! Documentation - TEST-NET-3
ip prefix-list BOGON-FILTER seq 70 deny 224.0.0.0/4 le 32    ! Multicast
ip prefix-list BOGON-FILTER seq 75 deny 233.252.0.0/24 le 32 ! Documentation - MCAST-TEST-NET
ip prefix-list BOGON-FILTER seq 80 deny 240.0.0.0/4 le 32    ! Reserved for future use
ip prefix-list BOGON-FILTER seq 90 permit 0.0.0.0/0 le 32    ! All address space
```

Then, in all the eBGP `neighbor` command we apply the filter to the input:

```
neighbor [neighbor ip address] prefix-list BOGON-FILTER in
```

In the router **BadGuy**, to simulate the advertising of `Bogons`, we created the `bogon` static routes pointing to `Null0` and in the BGP section we advertise them with the `redistribute static` command. This command allow the injection of all the static routes of the routing table in the BGP routing process

```
ip route 0.0.0.0 255.0.0.0 Null0      ! Software (current local network)
ip route 10.0.0.0 255.0.0.0 Null0      ! Private
ip route 100.64.0.0 255.192.0.0 Null0  ! Shared Address Space - CG-NAT
ip route 127.0.0.0 255.0.0.0 Null0     ! Loopback addresses to Localhost
ip route 169.254.0.0 255.255.0.0 Null0 ! APIPA - Automatic Private IP Address
ip route 172.16.0.0 255.240.0.0 Null0  ! Private
ip route 192.0.0.0 255.255.255.0 Null0 ! IETF Protocol Assignments
ip route 192.0.2.0 255.255.255.0 Null0 ! Documentation - TEST-NET-1
ip route 192.88.99.0 255.255.255.0 Null0 ! Reserved
ip route 192.168.0.0 255.255.0.0 Null0 ! Private
ip route 198.18.0.0 255.254.0.0 Null0  ! Benchmark testing
ip route 198.51.100.0 255.255.255.0 Null0 ! Documentation - TEST-NET-2
ip route 203.0.113.0 255.255.255.0 Null0 ! Documentation - TEST-NET-3
ip route 224.0.0.0 240.0.0.0 Null0     ! Multicast
ip route 233.252.0.0 255.255.255.0 Null0 ! Documentation - MCAST-TEST-NET
ip route 240.0.0.0 240.0.0.0 Null0     ! Reserved for future use
!
! =====
!                               BGP
! =====
!
router bgp 1
  bgp router-id 10.66.66.66
  !...
  redistribute static
```

Apply Remote Triggered Black Hole (RTBH) filtering to mitigate a DoS attack within AS1273

To simulate an attack from the Router **R14** to the **AS 1273** - Vodafone, we created the interface `Loopback 2` with the IP address `63.96.0.115 255.255.255.255`.

Inside the **AS 1273** - Vodafone, we configured the Router **R4** as a `trigger router` to filter the attacking prefix

```
interface GigabitEthernet1/0
  ! Internal OSPF link to R3 (10.3.4.0/30 network)
  ip address 10.3.4.2 255.255.255.252
  !uRPF (Unicast Reverse Path Forwarding)
  ip verify unicast reverse-path
  no shutdown
```



```

!
! =====
! Blackhole Routing Configuration (BGP Traffic Scrubbing / DDoS Mitigation)
! =====

! -----
! 1. Static Route to Null0 (Blackhole Route)
!   - Creates a discard route for a specific host (64.96.0.115/32)
!   - Traffic matching this route will be silently dropped by the router
!   - TAG 66: Used for route tracking or identification (can be matched in route-maps)
! -----
ip route 64.96.0.115 255.255.255.255 Null0 tag 66

! -----
! 2. Prefix List for Route Matching
!   - Defines the specific prefix (64.96.0.115/32) to be matched in route-maps
!   - Used to selectively apply BGP policies to this specific host route
! -----
ip prefix-list BH seq 5 permit 64.96.0.115/32

! -----
! 3. Route Map for BGP Announcement Manipulation
!   - Applied when advertising this prefix to BGP peers
!   - "black-hole-trigger": Descriptive name indicating DDoS mitigation trigger
!   - Sequence 10: First (and likely only) policy statement
! -----
route-map black-hole-trigger permit 10
! Match the specific blackhole prefix defined above
match ip address prefix-list BH

! Set BGP LOCAL_PREFERENCE to 200 (high priority within local AS)
! This makes the blackhole route preferred over regular paths
set local-preference 200

! Set BGP ORIGIN attribute to IGP (indicates internal origin)
set origin igp

! Add NO-EXPORT community (well-known community 0xFFFFF01)
! Prevents this route from being advertised outside the local AS
set community no-export

! Modify NEXT_HOP to a specific scrubbing center IP (192.0.2.1)
! This redirects traffic to a DDoS mitigation/scrubbing appliance
set ip next-hop 192.0.2.1

! -----
! 4. Static Route to Scrubbing Center
!   - Routes traffic destined to the scrubbing center (192.0.2.1) to Null0
!   - This is a LOOP PREVENTION mechanism:
!       1. BGP announces 64.96.0.115/32 with next-hop 192.0.2.1
!       2. Traffic to victim (64.96.0.115) goes to scrubbing center (192.0.2.1)
!       3. This route ensures traffic to scrubbing center itself doesn't loop
!       4. Actual scrubbing center reachability via other means (not shown)
! -----
ip route 192.0.2.1 255.255.255.255 Null0

! IN BGP SECTION
router bgp 1273
 redistribute static          ! Redistributes static routes (includes 64.96.0.115/32)
! ...
!-----
! CRITICAL: Apply blackhole route-map to outgoing advertisements
!   - Applies the 'black-hole-trigger' route-map to modify BGP attributes
!   - This ensures blackhole routes get special treatment when advertised
! -----
neighbor iBGP route-map black-hole-trigger out
! ...

```

5.3 - Test and Validation

Sessions in AS701 are in the Established state after applying the MD5 authentication

```
R14#sh ip bgp sum
BGP router identifier 10.14.14.14, local AS number 701
BGP table version is 40, main routing table version 40
28 network entries using 3696 bytes of memory
58 path entries using 3016 bytes of memory
20/9 BGP path/bestpath attribute entries using 3360 bytes of memory
14 BGP AS-PATH entries using 368 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 10472 total bytes of memory
BGP activity 33/5 prefixes, 63/5 paths, scan interval 60 secs
```

Neighbor	V	AS	MsgRcvd	MsgSent	TblVer	InQ	OutQ	Up/Down	State/PfxRcd
64.112.0.2	4	1273	11	13	40	0	0	00:02:08	17
64.112.0.6	4	17390	14	13	40	0	0	00:02:09	12
64.112.0.10	4	4637	11	13	40	0	0	00:01:58	19

Bogon prefixes from "BadGuy" do not enter the local BGP table in AS 1273

```
R14#sh ip bgp
BGP table version is 40, local router ID is 10.14.14.14
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
               r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network        Next Hop        Metric LocPrf Weight Path
*> 46.87.162.0/24  64.112.0.10          0 4637 5511 i
*                  64.112.0.2          0 1273 20717 5511 i
*                  64.112.0.6          0 17390 1273 20717 5511 i
*> 46.88.20.0/24   64.112.0.10          0 4637 5511 i
*                  64.112.0.2          0 1273 20717 5511 i
*                  64.112.0.6          0 17390 1273 20717 5511 i
*> 46.88.22.0/24   64.112.0.10          0 4637 5511 i
*                  64.112.0.2          0 1273 20717 5511 i
*                  64.112.0.6          0 17390 1273 20717 5511 i
*> 46.89.36.0/24   64.112.0.10          0 4637 5511 i
*                  64.112.0.2          0 1273 20717 5511 i
*                  64.112.0.6          0 17390 1273 20717 5511 i
*> 46.89.38.0/24   64.112.0.10          0 4637 5511 i
*                  64.112.0.2          0 1273 20717 5511 i
*                  64.112.0.6          0 17390 1273 20717 5511 i
*> 46.92.2.0/24    64.112.0.10          0 4637 5511 i
*                  64.112.0.2          0 1273 20717 5511 i
*                  64.112.0.6          0 17390 1273 20717 5511 i
* 48.73.239.0/24   64.112.0.10          0 4637 1273 i
*>                  64.112.0.2          0 1273 i
*                  64.112.0.6          0 17390 1273 i
* 48.73.240.0/22   64.112.0.10          0 4637 1273 i
*>                  64.112.0.2          0 1273 i
*                  64.112.0.6          0 17390 1273 i
*> 48.73.240.12/30  64.112.0.10          0 4637 i
* 63.25.64.0/18    64.112.0.10          0 4637 1273 i
*>                  64.112.0.2          0 1273 i
*                  64.112.0.6          0 17390 1273 i
*> 64.96.0.0/24    0.0.0.0              32768 i
*> 64.96.0.0/12    0.0.0.0              32768 i
s> 64.96.0.114/32  0.0.0.0              32768 i
s> 64.96.0.115/32  0.0.0.0              32768 i
s> 64.112.0.0/30    0.0.0.0              32768 i
*> 64.112.0.0/24    0.0.0.0              32768 i
*> 64.112.0.0/12    0.0.0.0              32768 i
s> 64.112.0.4/30    0.0.0.0              32768 i
s 64.112.0.8/30    64.112.0.10          0 4637 i
s>                  0.0.0.0              32768 i
*> 65.0.204.0/24   0.0.0.0              32768 i
* 130.41.46.0/24   64.112.0.10          0 4637 1273 17390 i
*                  64.112.0.2          0 1273 17390 i
*>                  64.112.0.6          0 17390 i
* 130.41.47.0/24   64.112.0.10          0 4637 1273 17390 i
*                  64.112.0.2          0 1273 17390 i
*>                  64.112.0.6          0 17390 i
```

* 158.23.228.0/24	64.112.0.6		0 17390 1273 5511 23344 i
*	64.112.0.10		0 4637 5511 23344 i
*>	64.112.0.2		0 1273 5511 23344 i
* 211.176.128.0/22	64.112.0.2		0 1273 4637 i
*>	64.112.0.10	0	0 4637 i
* 211.176.132.0	64.112.0.2		0 1273 4637 i
*>	64.112.0.10	0	0 4637 i
* 211.176.135.0	64.112.0.2		0 1273 4637 i
*>	64.112.0.10	0	0 4637 i
* 211.176.136.0/22	64.112.0.2		0 1273 4637 i
*>	64.112.0.10	0	0 4637 i
* 211.176.140.0/23	64.112.0.2		0 1273 4637 i
*>	64.112.0.10	0	0 4637 i

RTBH filtering

R14#ping 48.73.239.1 source lo2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 48.73.239.1, timeout is 2 seconds:
Packet sent with a source address of 64.96.0.115
.....
Success rate is 0 percent (0/5)

No.	Time	Source	Destination	Protocol	Length	Info
165	144.383927	64.96.0.115	48.73.239.1	ICMP	114	Echo (ping) request id=0x0000, seq=0/0, ttl=254 (no response found!)
166	146.461460	64.96.0.115	48.73.239.1	ICMP	114	Echo (ping) request id=0x0000, seq=1/256, ttl=254 (no response found!)
172	148.596688	64.96.0.115	48.73.239.1	ICMP	114	Echo (ping) request id=0x0000, seq=2/512, ttl=254 (no response found!)
175	150.817624	64.96.0.115	48.73.239.1	ICMP	114	Echo (ping) request id=0x0000, seq=3/768, ttl=254 (no response found!)
176	152.845675	64.96.0.115	48.73.239.1	ICMP	114	Echo (ping) request id=0x0000, seq=4/1024, ttl=254 (no response found!)

Capture of ICMP traffic between R5 and R3, pinging (ICMP request) from R14-Lo2 to Server 2

5.4 - Practical Questions

5.4.1 - How does MD5 authentication improve the security of BGP sessions? What happens if it is not enabled?

1. Protection Against Session Hijacking

MD5 authentication prevents unauthorized routers from establishing BGP sessions with your router by requiring a shared secret key.

2. Defense Against Replay Attacks

The MD5 hash includes a sequence number/timestamp, making it resistant to replay attacks where an attacker captures and resends valid BGP packets.

3. Prevention of Rogue Peer Establishment

Without authentication, any router claiming to be from a legitimate AS could form a BGP session and:

- Inject malicious routes
- Hijack prefixes
- Cause routing instability

4. Integrity Verification

Each BGP packet is checksummed with the secret key, ensuring the packet hasn't been tampered with in transit.

5. Basic Peer Identity Verification

While not strong cryptographic authentication, MD5 provides a basic level of identity verification for BGP peers.

What Happens Without MD5 Authentication

Critical Vulnerabilities Exposed:

1. Route Hijacking Attacks

```
# Example of what an attacker could do:
# 1. Spoof BGP OPEN message from legitimate peer
# 2. Establish unauthorized BGP session
# 3. Announce hijacked prefixes:
network 8.8.8.0/24    # Hijack Google DNS
network 203.0.113.0/24 # Announce bogon space
```

Real-World Example: In 2018, attackers hijacked Amazon's Route 53 DNS prefixes, redirecting traffic to malicious servers. [link](#)

2. BGP Session Reset Attacks

Without authentication, attackers can send:

- **TCP RST packets** to tear down legitimate BGP sessions
- **BGP NOTIFICATION messages** to cause session resets
- **Malformed BGP packets** to crash BGP processes

3. Route Poisoning/Instability

Attackers can inject:

- **Malformed attributes** (corrupted AS_PATH, communities)
- **Excessive updates** causing CPU exhaustion
- **Withdrawals for critical prefixes** causing blackholes

4. Man-in-the-Middle Attacks

Legitimate Path: AS1273 <--> [Internet] <--> AS5511

Compromised Path: AS1273 <--> [Attacker] <--> AS5511

Attacker can:

1. Intercept and modify BGP updates
2. Learn internal network topology
3. Perform traffic analysis
4. Selectively drop or modify routes

5. Prefix De-aggregation Attacks

Legitimate announcement: 192.0.2.0/24

Attacker announcement: 192.0.2.0/25

192.0.2.128/25

More specific routes win, allowing traffic hijacking.

Impact of Missing Authentication - Case Study

The YouTube Hijack Incident (2008)

WHAT HAPPENED:

- Pakistan Telecom (AS17557) announced YouTube's prefix (208.65.153.0/24)
- No authentication between Pakistan Telecom and PCCW (AS3491)
- Route propagated globally
- YouTube inaccessible worldwide for ~2 hours

ROOT CAUSE:

1. No BGP session authentication
2. No route filtering
3. No RPKI/ROV validation
4. No maximum prefix limits

Financial Impact:

- **Direct costs:** \$1.5M+ in lost ad revenue (YouTube)
- **Indirect costs:** Brand damage, customer trust erosion
- **Operational costs:** Emergency response, mitigation efforts

[Wikipedia page about Google Outages](#)

Without MD5 authentication, BGP sessions are vulnerable to:

- ✓ Session hijacking
- ✓ Route poisoning
- ✓ Traffic interception
- ✓ Denial of service
- ✓ Prefix hijacking

5.4.2 - Which Bogon ranges did you filter in your lab, and why must they not appear in the global routing table?

Filtered Ranges:

Private/Internal Space:

- 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16 - Corporate/Home networks
- 100.64.0.0/10 - Carrier-Grade NAT (ISP internal use)

Special/Reserved:

- 0.0.0.0/8 - "This" network
- 127.0.0.0/8 - Loopback (localhost)
- 169.254.0.0/16 - Link-local auto-configuration

Documentation/Testing:

- 192.0.2.0/24, 198.51.100.0/24, 203.0.113.0/24 - Test networks
- 233.252.0.0/24 - Multicast testing

Reasons why must they not appear in the global routing table

1. Security Threats:

- Spoofing attacks - Fake sources enable DDoS
- Route hijacking - Redirect internal traffic
- Resource exhaustion - Fill routing tables with junk

2. Breaks Networks:

- VPNs fail when private routes leak
- NAT stops working
- Cloud services conflict

3. Violates Standards:

- RFC 1918: Private addresses must stay local
- BCP 38: Required ingress filtering
- MANRS: Industry security requirement

4. Internet Stability:

- Pollutes global BGP table (~900K routes)
- Causes routing confusion globally
- Wastes ISP resources

5.4.3 - Explain how RTBH was implemented in AS1273. Include a diagram showing how the trigger router propagates the blackhole route

RTBH was implemented in **AS1273 (Vodafone)** using a **trigger-based approach** where **Router R4** acts as the **trigger router** that injects and propagates the blackhole route throughout the AS.

1. Attack Identification & Static Route Creation

```
! On R4 (Trigger Router)
ip route 64.96.0.115 255.255.255.255 Null0 tag 66
```

- **Purpose:** Creates a discard route for the attacking IP (64.96.0.115/32)
- **Action:** Traffic matching this route is silently dropped at R4
- **Tag 66:** Used for route identification in redistribution policies

2. Route Redistribution into BGP

```
router bgp 1273
 redistribute static
```

- **Purpose:** Injects the static blackhole route into BGP
- **Result:** The /32 host route becomes a BGP prefix that can be advertised

3. BGP Attribute Manipulation via Route-Map

```
! Prefix list to match the attack target
ip prefix-list BH seq 5 permit 64.96.0.115/32

! Route-map to modify BGP attributes
route-map black-hole-trigger permit 10
 match ip address prefix-list BH
 set local-preference 200          ! Make it preferred internally
 set origin igp                   ! Mark as internally originated
 set community no-export          ! Prevent export outside AS1273
 set ip next-hop 192.0.2.1        ! Redirect to scrubbing center

! Apply route-map to iBGP peers
neighbor iBGP route-map black-hole-trigger out
```

Key Attribute Settings:

- **LOCAL_PREF 200:** Ensures all iBGP peers prefer this route over any external paths
- **NO-EXPORT Community:** Prevents the route from being advertised to eBGP peers
- **Next-Hop 192.0.2.1:** Points to a (simulated) scrubbing/drop center

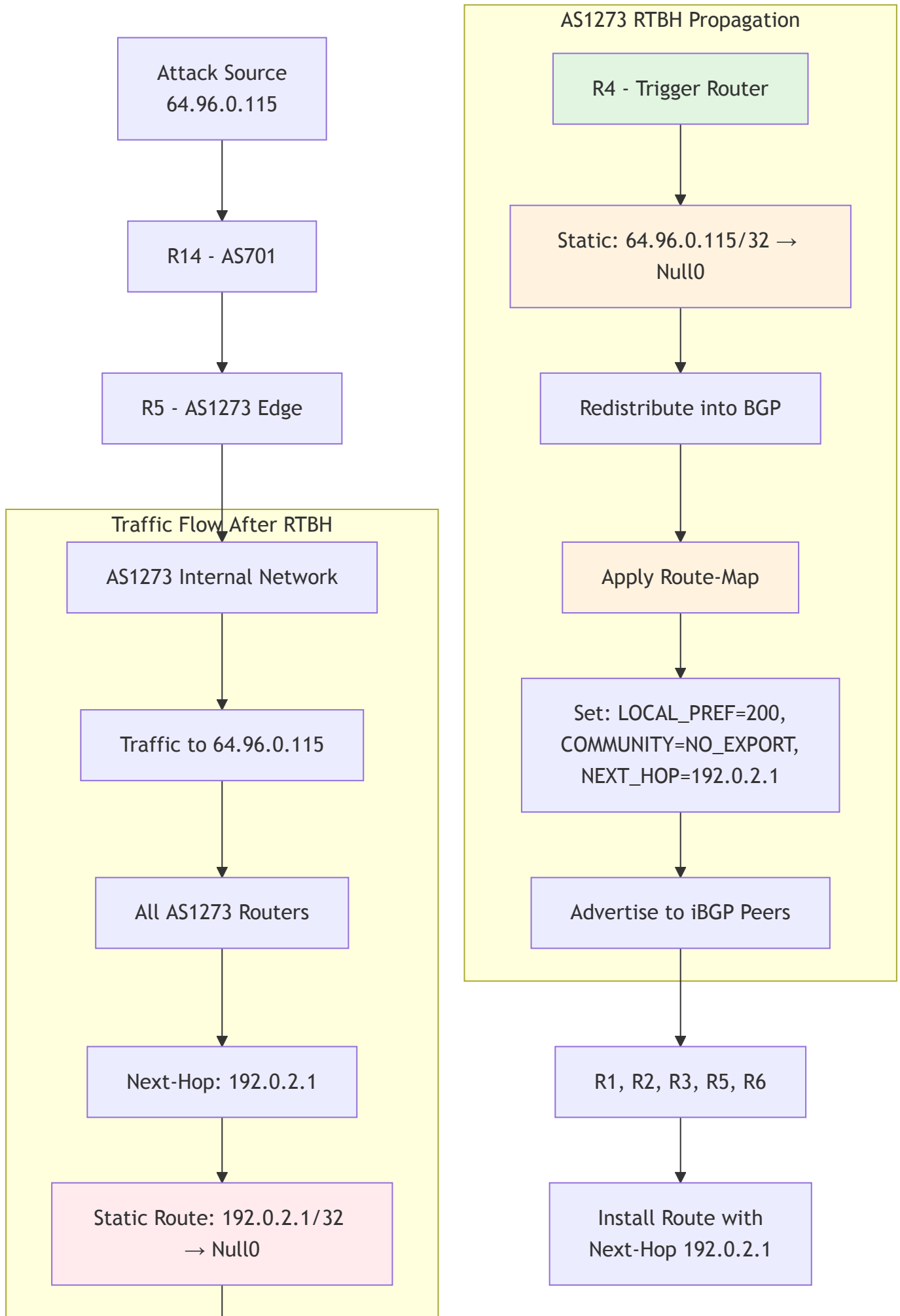
4. Scrubbing Center Route & uRPF

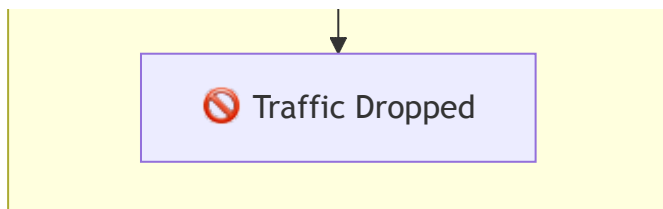
```
! Static route for the scrubbing center
ip route 192.0.2.1 255.255.255.255 Null0

! uRPF on ingress interfaces
interface GigabitEthernet1/0
 ip verify unicast reverse-path
```

- **192.0.2.1:** Reserved TEST-NET address used as the scrubbing destination
- **uRPF:** Validates source addresses and prevents spoofing

5. Propagation Throughout AS1273





How the Blackhole Route Propagates

Step 1: Trigger Injection

- **R4** creates the blackhole static route with tag 66
- Route is redistributed into BGP with `redistribute static`

Step 2: Attribute Transformation

- The route-map `black-hole-trigger` matches the prefix (`64.96.0.115/32`)
- BGP attributes are modified:
 - **LOCAL_PREF**: Increased to 200 (default is 100)
 - **COMMUNITY**: `NO_EXPORT` added
 - **NEXT_HOP**: Changed to `192.0.2.1`
 - **ORIGIN**: Set to `IGP`

Step 3: Internal Propagation

- Modified route is advertised to all iBGP peers (R1, R2, R3, R5, R6)
- Due to **LOCAL_PREF=200**, all routers prefer this internal route over any external paths
- **NO_EXPORT** community prevents advertisement to eBGP peers (AS701, AS17390, etc.)

Step 4: Traffic Redirection

1. **Incoming attack traffic** enters AS1273 via R5 or R6
2. All routers have the best path: `64.96.0.115/32 → Next-Hop 192.0.2.1`
3. Routers forward traffic toward `192.0.2.1`
4. **Static route on all routers**: `192.0.2.1/32 → Null0`
5. **Result**: Traffic is silently dropped at the ingress point to the scrubbing center

Step 5: uRPF Validation

- **uRPF** on edge interfaces ensures spoofed return traffic is dropped
- Prevents the attacker from using spoofed source addresses

Key Design Principles

1. **Centralized Trigger**: Single point (R4) controls blackhole activation
2. **Selective Propagation**: Only affects the specific attack target (`/32`)
3. **Containment**: `NO_EXPORT` keeps the blackhole within AS1273
4. **Scalability**: Can trigger multiple blackholes simultaneously
5. **Reversibility**: Simply remove the static route on R4 to restore normal routing

This implementation provides a **rapid, scalable DDoS mitigation** mechanism that can drop attack traffic within seconds of activation, while maintaining normal service for all other traffic.

5.4.4 - Describe the role of uRPF in validating traffic source addresses and preventing spoofing

Core Function:

- Validates that incoming packets arrive on the interface that would be used to send packets BACK to the source address.

Benefits:

1. Stops DDoS Reflection/Amplification

Common DDoS: Spoof victim's IP → DNS/NTP servers → Amplified reply floods victim
With uRPF: Spoofed packets dropped at first hop

2. Prevents IP Address Hijacking

- Can't impersonate other networks
- Can't use bogon addresses
- Can't spoof internal resources

uRPF acts as a "return address checker" for network traffic, dropping any packets that arrive on the "wrong door" based on our routing table, making source address spoofing virtually impossible at the network edge and in our case to enable the RTBH.

5.4.5 - What impact would the attack from 64.96.0.115 have on AS1273 if RTBH was not deployed?

Immediate Technical Impact:

- **Link Saturation:** All Vodafone uplinks (R5→AS701, R6→AS4637, R6→AS5511) become 100% saturated
- **Router Resource Exhaustion:** Border routers (R5, R6) hit 100% CPU, memory exhaustion
- **Service Blackout:** All Vodafone customers lose Internet connectivity
- **BGP/OSPF Instability:** Routing protocols crash due to resource exhaustion
- **Business Impact::** Revenue loss, SLA violations, customer churn and reputation damage
- **Long response Time Without RTBH**
- **Cascade Effects:**
 - **Backscatter:** Innocent victims receive attack responses
 - **Peering Impact:** Neighbors (AS5511, AS4637) experience congestion
 - **Blacklisting:** AS1273 potentially gets blacklisted by DDoS protection services

Conclusion

The completion of **Lab Project No. 2 - BGP Routing Configurations** has provided us with an end-to-end, practical understanding of inter-domain routing in the Internet. This hands-on journey from basic IGP configuration to advanced BGP policy implementation and security hardening has been instrumental in solidifying the theoretical concepts covered in the Internet Networks course.

Throughout the five phases, we successfully:

1. **Built a Robust Foundation:** Established a stable OSPF backbone within each AS, including a complex multi-area design with a stub area in AS 5511, ensuring efficient internal routing.
2. **Engineered Inter-AS Connectivity:** Configured both iBGP (using route reflectors for scalability) and eBGP sessions to seamlessly connect all eight Autonomous Systems, enabling global reachability as validated by comprehensive TCLSH ping tests.
3. **Implemented Professional-Grade Policies:** Applied real-world routing policies, including prefix aggregation, filtering of private AS numbers and long prefixes (> /24), and influencing path selection using attributes like weight and local preference to meet specific traffic engineering goals.
4. **Fortified the Network:** Enhanced the security posture of our topology by implementing MD5 authentication for BGP peers, deploying strict bogon prefix filters at network edges, and configuring a sophisticated RTBH system to mitigate potential DDoS attacks.

The key takeaways from this project extend beyond specific commands. We have learned the critical importance of **hierarchical design** (OSPF areas, route reflection), the **policy-centric nature of BGP**, and the **non-negotiable requirement for security** in inter-domain routing.

This lab has effectively showed us the operational reality of large-scale networks. The skills developed—encompassing design, configuration, troubleshooting, and policy implementation—are directly applicable to roles in network engineering, ISP operations, and cloud infrastructure. We conclude this project with a significantly deepened appreciation for the complexity and elegance of the protocols that keep the global Internet connected and secure.